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Revisiting the traffic flow observability problem: A matrix-based model for traffic networks with or without centroid nodes

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ABSTRACT

This study introduces a graph theory-based model that addresses the link flow observability problem in traffic networks by optimizing passive sensor deployment. The model aims to determine the minimal number of sensors and their optimal placement. It constructs a virtual network and uses isomorphic graph theory to map between the original and virtual networks, ensuring consistency in nodes, links, and link directions. Two formulas are proposed to calculate the minimum number of observable links required across different networks, factoring in links, ordinary nodes, centroid nodes, and added links. Key concepts such as chords, cut sets, and loops, along with their matrices, are analyzed. A matrix-based framework is developed to consider flow conservation conditions. Results show that solving the full link flow observability problem using node flow conservation equations yields a fixed number of sensors with non-unique deployment schemes, Additionally, a resource-constrained sensor network optimization (RSNO) model is presented, employing null space projection (NSP) as an objective function to quantify the impact of budget constraints particularly under the condition if all the link flows cannot be observed. Numerical examples demonstrate the RSNO model's applications.

1. Introduction

In contemporary urban transportation systems, the precise measurement and analysis of traffic flow play a pivotal role in traffic planning and management. These data are not only crucial for assessing traffic demand and guiding urban road design, but also important to enhance traffic safety and operational efficiency. Technological advancements have enabled advanced sensor systems to automatically collect traffic flow data, providing robust support for traffic planning and management. These systems facilitate real-time monitoring of traffic flow by directly observing link flow, and path flow in certain cases, or indirectly monitoring aggregated flow such as origin-destination (OD) travel counts (Xu et al., 2023).

In traffic networks, the relationships between flows are typically represented by a system of linear equations. In this system, the columns of the coefficient matrix represent number of the unknown link flows, while the rows relate to the data collected by the deployed sensors. These sets of equations define a model directly linked to sensor deployment, with the existence and uniqueness of their solutions depending on the type and number of sensors. This gives rise to two key research questions (Gentili and Mirchandani,

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2012): (1) determining the minimum number of sensors required for obtaining a unique solution for the set of equations along with their optimal locations; (2) selecting appropriate sensor locations to improve the quality of flow estimation if the set of equations does not yield a unique solution. These two questions further motivate research progress on sensor deployment based on observability and state estimation-based traffic state detection.

In research focused on observability, the key objective is to determine whether a specific subset of observed traffic flow values (collected by deployed sensors) is adequate for estimating another given subset of observable traffic flow values (Castillo, 2012). The term "observable" specifically refers to those traffic flow values that, although not directly measured, can be inferred from existing observed data. Observability in this context pertains to the information obtainable from the general solution of a system of linear equations and/or inequalities (Castillo, 2001; 2002).

In this study, passive sensors are placed on particular links to utilize traffic flow observation data from these links to infer the traffic conditions of links without sensors. To differentiate between these two types of links, we adopt the following terminologies: links with deployed sensors are referred to as "observed links", while those without sensors are termed "unobserved links". Maximizing the utilization of advanced sensor systems hinges on effectively deploying sensors within the traffic network to optimize observability and estimation accuracy. This encompasses two main research areas: full observability and partial observability.

Full observability refers to the condition where all states are observable based on data measured by available detectors (Gentili and Mirchandani, 2014). The observability problems considered in this category of research include link flow observability, path flow observability, and OD flow observability (Castillo, 2013b; Liu and Zhou, 2019; Owais, 2022). Partial observability, on the other hand, focuses on maximizing observable states under limited sensor resources (Gentili and Mirchandani, 2012; Castillo, 2012; Ng, 2013; Viti, 2012, 2014). Observability spans from full observability to full unobservability, with partial observability involving the determination of only some unknown variables and deriving bounds for the remaining variables.

In this paper, the research on sensor location based on observability is divided into (1) Sensor location problems for link flow inference/observability; (2) Sensor location problems for path flow observability/path reconstruction; and (3) Sensor location problems for OD estimation/updating. There exist several approaches (see Fig. 1 for a summarized view) to address specific forms of the general sensor location flow observability problem.

1.1. *Sensor location problems for link flow inference/observability*

The widely concerned/early investigated observability problem is the link observability problem based on link flow, which involves directly observing traffic flow on links with sensors. The traffic flow data on the links in a transportation network provides valuable information for addressing long-term planning and short-term operational needs. For example, link traffic flow can be used to infer OD demand and evaluate the level of service (Highway Capacity Manual, 2000).

Researchers have addressed the link flow inference/observability problem by optimally locating passive sensors on the links. Gu (2005) introduced the concept of a minimum link control set for sensor localization, while Chin et al. (2009) demonstrated the NP-hard nature of sensor placement on undirected graphs and provided polynomial approximation algorithms. He (2013) transformed the problem into a minimum-spanning tree search in a directed graph.

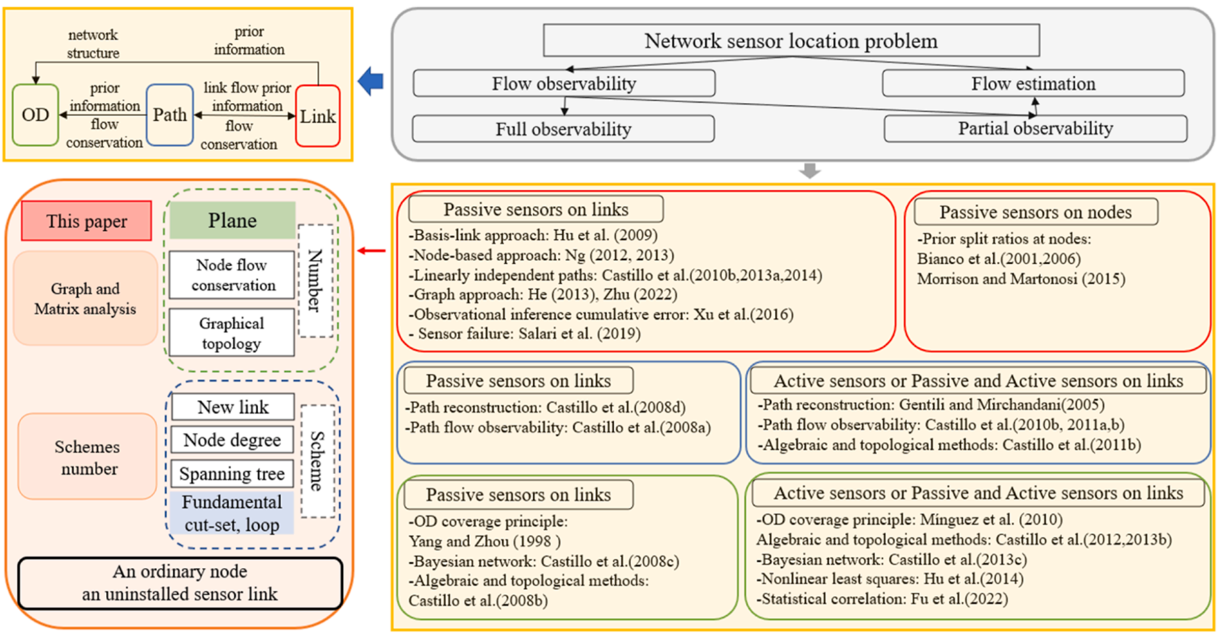


Fig. 1. Research on sensor location based on observability.

Hu et al. (2009) and Ng (2012) developed sensor placement methods that utilized matrix and flow conservation to minimize the number of sensors required in a traffic network. The difference lies in Hu et al.'s assumption that all path information is known and their use of a path-link incidence matrix, while Ng (2012) used a node-link incidence matrix to avoid section enumeration. This work was further extended to address the link part observability problem (Ng, 2013).

Castillo et al. (2013a, E. 2014) refined these methods by focusing on minimal path vectors and identifying network "holes". Zhu (2021) advanced this field with a mixed integer optimization model for full observability based on network topology.

In using mixed sensors deployed on links, Xu et al. (2016) introduced "new links" and a robust model accounting for measurement error propagation, while Salari et al. (2019) proposed a predictive model using two sensor types for different levels of link importance.

The problem of positioning passive sensors on network nodes to achieve full observability of link flows was firstly addressed by Bianco et al. (2001). Subsequently, Bianco et al. (2006, L. 2014) researched the placement of minimal sensors on nodes for traffic flow observability, utilizing turning ratios and conservation equations. They tackled the NP-hard problem using combinatorial methods such as branch-and-bound and genetic algorithms. Morrison and Martonosi (2015) corrected a theorem from Bianco et al. (2001) and provided a stronger necessary condition for symmetric directed trees.

1.2. Sensor location problems for path flow observability/path reconstruction

In the context of traffic flow monitoring, while sensors can capture link flows at specific locations, they may not provide information on the paths taken by users. Path flow observability aims to observe all path flows and reconstruct the complete picture of paths used between origin-destination pairs using captured flow information. Methods such as combining optimization attributes or utilizing license plate scanning recognition detection technology can be employed to address path flow observability.

The concept of full and partial observability of path flows through active sensor placement on network links was initially studied by Castillo et al. (2008d, 2010a), who presented a mathematical model for the problem. Subsequent research by Castillo et al. (2010b) focused on optimizing camera placement for maximum path flow detection and efficient scanning resource utilization in traffic network path flow estimation, addressing both full and partial observability problems.

Cerrone et al. (2015) addressed the full and estimation versions of the traffic observability problem in mixed sensor deployment scenarios, balancing sensor cost with the need for complete path flow volume derivation under budget constraints, and later introduced a mix of active and passive sensors to maintain path quality. Fu et al. (2016) introduced a strategic model for combining fixed and mobile license plate sensors to optimize path reconstruction and sensor deployment costs, while Shan et al. (2018) developed a fully observable two-layer model for deploying a mix of scanning and counting sensors in a network.

1.3. Sensor location problems for OD estimation/updating

OD matrix estimating/updating is one of the earliest applications of traffic flow data (Gaudry and Lamarre, 1979). Research on OD flow observability problems is common, covering both link flow as well as path flow estimation. All of these methods depend on the availability of extra information as priori information (e.g., historical OD matrix), where the reliability of such prior information is vital. These existing studies can be reviewed from the following three aspects.

Regarding the passive sensors deployed on links, Yang and Zhou (1998) established fundamental rules for sensor placement to maximize coverage of origin-destination (OD) traffic. Castillo et al. (2008c) employed a Bayesian network model to estimate traffic flows and update OD matrices. Li and Ouyang (2011) developed a sensor deployment model based on vehicle ID detection for OD estimation. Salari et al. (2021) presented a model for identifying reliable sensor locations, while Sun et al. (2021) proposed a method to evaluate OD demand estimation error and developed a bi-objective optimization model aimed at minimizing errors in OD mean and covariance estimation.

For active sensors deployed on links, Mínguez et al. (2010) investigated the optimal placement of license plate scanners for OD estimation under various constraints, while Hadavi and Shafahi (2016) utilized vehicle recognition data for OD estimation and sensor localization. Sun et al. (2022) introduced an AVI sensor model that prioritizes OD demands and accounts for sensor failures to ensure the unique identification of OD demand without path flow determination.

Finally, some researchers investigated the problem of mixed sensors deployed on links. Castillo et al. (2012, 2013b) worked on processing license plate data to address the traffic flow observability problem related to both OD and path flow using matrix techniques. Hu and Liou (2014) developed an integrated fixed-mobile detector data-based method for estimating od flow, formulating it as a non-linear least squares problem with linear constraints. Fu et al. (2022) introduced a model optimizing senor placement to reduce uncertainty in OD demand estimation considering covariance across periods.

However, few of these existing studies considered the total number of sensor deployment schemes. In this paper, a matrix-based method is proposed to give the specific sensor deployment scheme and the total number of sensor deployment schemes using the information of the spanning tree.

The rest of this paper is organized as follows. Section 2 outlines the research motivation and contributions; Section 3 introduces the definitions and theorems necessary for understanding the paper; Section 4 presents an algorithm for obtaining a virtual network; Section 5 proposes a method for deriving the fully observable link problem from the virtual network, explicitly calculating the minimum number of required sensors; Section 6 utilizes graph theory and matrix analysis for node flow conservation; Section 7 introduces the resource-constrained sensor network optimization (RSNO) model, designed to enhance traffic network performance under budget constraints; and Section 8 provides the conclusions of this paper.

2. Motivation and contribution

Mathematically, the existing studies on traffic flow observability problems can be classified into three categories, optimization-based models, algebraic-based models, and graph theory-based models.

(1) Optimization-based models. Intuitively, the traffic flow observability problem can be regarded as minimizing the number of sensors or other objectives, using observed data from observable links equipped with sensors, to calculate traffic flow in unobserved links (without sensor deployment). Such kind of models usually require some data sources, commonly including counting, scanning, or “prior” data, the most common constraints include conservation laws, flow nonnegativity, link capacity, flow definition, observation, flow propagation, and specific model requirements (Castillo et al., 2015).
(2) Algebraic-based models. Castillo et al. (2001, 2002) pointed out that the observability problem corresponds to an algebraic problem, as the conservation constraints of traffic flow can be expressed as a system of linear equations and non-negativity inequalities, making observability equivalent to the algebraic compatibility of these linear systems. For example, Castillo et al. (2007; 2010b) proposed that general flow observable problems can be solved with linear algebraic theories by using node-based flow conservation relations, and provided matrix tools. Hu et al. (2009) first attempted to identify the smallest subset of links in a network and proposed an algebraic basis connection method to determine the position of vehicle sensors, using the link-path incidence matrix to represent the network structure.
(3) Graph theory-based models. The traffic flow observability problem has a close relationship with the network topology. Thus, the graph theories are helpful to address the traffic flow observability problem. For example, He (2013) solved the link flow observability problem using spanning trees without considering paths. However, if all the nodes in a network are centroid nodes, such a method is not applicable.

Hu (2014) found that the deployment conditions for passive sensors on links are stricter than those for active sensors, based on node flow conservation, yet their upper bounds are the same. This suggests that the deployment model for passive sensors can be applied to the deployment of active sensors, but the locations suitable for active sensors may not necessarily be suitable for passive sensors. Building upon the concept of a plane in graph theory as defined by Castillo et al. (2014), this study utilizes algebraic methods and graph theory to deploy passive sensors on links, address the link flow inference/observability problem, and re-evaluate the traffic flow observability problem. For simplicity, throughout the paper, the term "sensor" denotes the "passive sensor" unless otherwise specified.

In this paper, the proposed method only requires observed link flows as input, without needing other flow information such as node turning coefficient (Bianco et al., 2001; 2006; 2014), all path information (Hu et al., 2009), and OD matrices (Mínguez et al., 2010; Sun et al., 2022). The output of this model is to determine the number and locations of links where sensors need to be deployed, which aligns with existing studies on link flow observability problems (Hu et al., 2009; Ng, 2012; He, 2013; Castillo et al., 2014).

The main contributions of this paper are as follows.

(1) The concepts of the plane and isomorphic graph used in graph theory are introduced following Castillo et al. (2014). Full use of the topology of traffic networks is made to investigate the link observability problem. Simple formulas are given for the numbers of links, ordinary nodes, virtual network planes, added links, and centroid nodes to determine the upper bound associated with the node-based traffic conservation equation, which is used to calculate the number of links required for the full link flow observability problem.
(2) A generalized model is given using the concepts of the plane and isomorphic graph used in graph theory. The number and locations of sensor links in the traffic network that need to be developed to solve the problem of a fully observable link flow are obtained directly through the proposed transformation steps of the virtual network.
(3) Concepts such as chords, cut-sets, and loops with their relevant metrics are employed in this study to use more information on traffic networks. Consequently, the fundamental incidence matrix, fundamental cut-set matrix, fundamental loop matrix, and plane matrix are used to describe relationships between links and nodes in a traffic network. Based on the aforementioned concepts, a matrix-based framework is proposed to account for the flow conservation conditions. This framework demonstrates that while the total number of sensors needed for full link flow observability remains constant, the number of possible deployment schemes is not unique. This variability is reflected in the number of spanning trees in the network, indicating that multiple deployment schemes are feasible.
(4) Establishes the resource-constrained sensor network optimization (RSNO) model, and uses null space projection (NSP) metrics to accurately assess the impact of partial observability on traffic network performance under budget constraints.

3. Concepts and network types

The full link flow observability problem refers to identify the minimum subset of links equipped with sensors that can guarantee the full observation of the all link flows in the network. This means that, based on the observed link flow from sensors on some links, the flows on the remaining links without sensors can be accurately inferred through a system of linear equations. To fully understand the concept of full observability, it is important to define some related terms.

3.1. Related definitions and concepts

Definition 1. **(Graph)** The general traffic network is represented by a directed graph $G = (N, L)$, where N is a node set, and L is a link set. The order of a graph is the number of nodes of the graph; i.e., a graph of order n has n nodes. The nodes of the traffic network are further divided into centroid and ordinary nodes. The centroid node refers to the OD node of the flow in the traffic network, whereas ordinary nodes refer to nodes other than the centroid node in the traffic network. N^* represents the centroid node set. Assuming that $|N| = n$, $|L| = l$, and $|N^*| = c$, the number of ordinary nodes is $(n - c)$, where generally $n < l$.

Definition 2. **(Isomorphic graph)** An isomorphic graph assumes that $G_1 = (N_1, L_1)$ and $G_2 = (N_2, L_2)$, if there is a bijection $\rho: N_1 \to N_2$, then all $u_i, u_j \in N_1$, $(u_i, u_j) \in L_1$, which is equivalent to $(\rho(u_i), \rho(u_j)) \in L_2$, G_1 and G_2 are isomorphic. Such a mapping ρ is called isomorphic mapping. Here, $(u_i, u_j) \in L$ denotes that (u_i, u_j) is a link and the node u_i is adjacent to and points to the node u_j. That is, the isomorphic mapping indicates that the adjacency of nodes u_i and u_j in the graph G_1 is equivalent to the adjacency of nodes $\rho(u_i)$ and $\rho(u_j)$ in the graph G_2.

For isomorphic graphs, the numbers of nodes and links are the same, and the connection directions of the links are the same. Isomorphism is an equivalence relation. Isomorphic graphs have the same order, but graphs of the same order are not necessarily isomorphic.

Proposition 1. *Set $G_1 = (N_1, L_1)$ and $G_2 = (N_2, L_2)$ as two simple graphs of order n. $N(G_1) = \{\mu_1, \mu_2, \cdots, \mu_n\}$, $N(G_2) = \{u_1, u_2, \cdots, u_n\}$, and their incidence matrices are an $n \times l$-order matrix A. Then, $G_1 \cong G_2$.*

Proof: Mapping

$$\rho : G_1 \to G_2, \\ \rho(\mu_i) \to u_i, \quad i = 1 \cdots n$$

ρ is a bijection.

If adjacent to v_i and v_j, and their connection has a directed link (μ_i, μ_j), then

$$a_{\mu_k(\mu_i,\mu_j)} = \begin{cases} -1 & \mu_k = \mu_i \\ 1 & \mu_k = \mu_j \\ 0 & otherwise \end{cases} \tag{1}$$

The incidence matrix G_2 is also A, and u_i and u_j are thus adjacent; i.e., $\rho(u_i) \sim \rho(u_j)$.

Conversely, if adjacent to u_i and u_j, i.e., $\rho(u_i) \sim \rho(u_j)$, we know G_1 and G_2 incidence matrices are an $n \times l$-order matrix A, their connection has a directed link (μ_i, μ_j) where the elements in matrix A are equal, we also have $\mu_i \sim \mu_j$.

In summary, ρ is an isomorphic mapping from the graph G_1 to graph G_2. Therefore, $G_1 \cong G_2$.

A simple example of graph isomorphism is presented in Fig. 2.

To analyze the topological structure of the traffic network, we introduce the concept of planar graphs and planes from graph theory to the traffic network.

Definition 3. **(Planar graph)** A graph G is considered a planar graph, if it can be drawn on a plane with no links crossing each other except at nodes. Any other type of network is classified as non-planar.

Simple examples of a planar graph and a non-planar graph are presented in Fig. 3a and b respectively.

Definition 4 (Added link) A planar graph has links that do not intersect except at its nodes. The addition of parallel links or loops does not change its planarity. However, when a link is added between non-adjacent nodes, the previously planar network becomes non-planar due to intersecting links. This additional link is referred to as an added link.

The non-planar network shown in Fig. 3b is compared with the planar network depicted in Fig. 3a. Link (2,6) connecting nodes 2 and 6 and Link (3,4) connecting nodes 3 and 4 (represented by black dotted lines) are added links. Link (2,6) intersects with link (3,5), while link (3,4) intersects with link (2,5), and also intersects with link (2,6). Additionally, there are solid black lines forming rings at node 8′s links (8,8). Furthermore, there are two solid black lines connecting node 5′s links (5,4) and node 4′s links (5,6) forming parallel links. Link (1,7), link (8,8), link (5,4) and link (5,6) are not added links because they do not intersect with any other links.

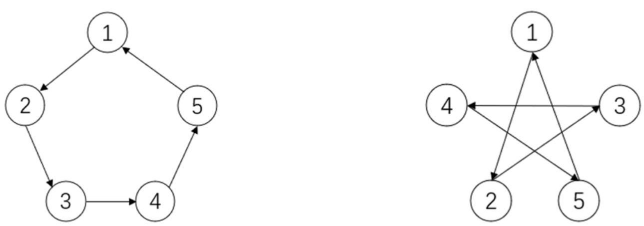


Fig. 2. Graph isomorphism.

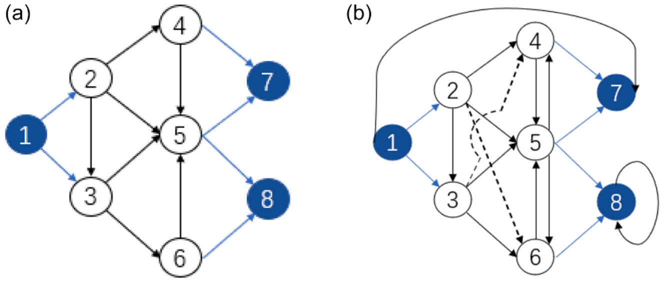


Fig. 3. (a) Planar graph of a small traffic network. (b) Non-planar graph of a small traffic network.

Definition 5. **(Network plane)** The links of a planar network G divide the plane into several areas, with each area being referred to as a plane. An outer plane is infinite in size, while an inner plane is finite in size. The links comprising the network plane are subsets of all links and consist of undirected loops that do not contain another plane inside.

A finite plane is an area within a ring (e.g., the starting and ending points of the ring are nodes 8). A finite plane can also be the area encircled by some links that connect two or more adjacent nodes, such as the area between nodes 3 and 4, or the area encircled by the links between nodes 1, 3, and 4). Adjacent nodes refer to two directly connected nodes, such as nodes 3 and 6.

A simple example of a network plane is presented in Fig. 4.

3.2. Relationship to existing full observability studies

Define a traffic network $G = (N, L)$ having n nodes and l links. In the traffic network, a node follows the flow conservation equation:

$$AV = \sum_{e\in O_u} v_e - \sum_{e\in D_u} v_e \tag{2}$$

where e is the link; v_e is the traffic flow on the link e; and O_u and D_u are the flow outflow link set and the flow inflow link set at node u, respectively.

A is the node–link incidence matrix, and its elements only take the values 0, 1, and -1:

$$a_{ue} = \begin{cases} 1 & e \in O_u \\ -1 & e \in D_u \\ 0 & otherwise \end{cases} \tag{3}$$

If a link e is connected to node u and the flow of link e goes to node u, then $a_{ue} = 1$.

If a link e is connected to node u and the flow of the link e comes from node u, then $a_{ue} = -1$.

If a link e is not connected to a node u, $a_{ue} = 0$.

For $u \in N \backslash N^*$, the centroid nodes, have $\sum_{e\in O_u} v_e - \sum_{e\in D_u} v_e = s_u$, where s_u is the equilibrium flow at a node u. For $u \in N^*$, the ordinary node, has $\sum_{e\in O_u} v_e - \sum_{e\in D_u} v_e = 0$.

Considering that all nodes in the traffic network are ordinary nodes, for any node, the total flow into the node equals the total flow out of the node, and the node flow conservation equation $AV = \sum_{e\in O_u} v_e - \sum_{e\in D_u} v_e = 0$ is rewritten as linear equations:

$$A_{n\times l} V_{l\times 1} = 0 \rightarrow \begin{cases} a_{11}v_1 + a_{12}v_1 + \ldots + a_{1l}v_1 = 0 \\ a_{21}v_1 + a_{22}v_1 + \ldots + a_{2l}v_1 = 0 \\ \vdots \\ a_{n1}v_1 + a_{n2}v_1 + \ldots + a_{nl}v_1 = 0 \end{cases} \tag{4}$$

$A_{n\times l}$ represents a node-link incidence matrix for a traffic network, while $V_{l\times 1}$ denotes the vector of link flow. The number of

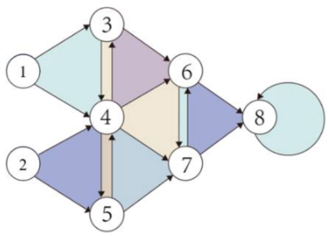


Fig. 4. Illustration of planar network division into finite and infinite planes.

equations in Eq. (4) corresponds to the number of nodes n, and the number of unknowns equals the number of links l. The linear equations have a unique solution when there are an equal number of equations and unknown variables which are all linearly independent. Therefore, in Eq. (4), it is possible to determine up to n unknown variables at most. When there are known remaining variables $(l-n)$, all n equations in the system are linearly independent resulting in Eq. (1) having a unique solution.

The following theorem provides an upper bound on observed links required for achieving full observability of link flow in a traffic network when linear equations have a unique solution.

Theorem 1. *It is assumed that traffic network $G=(N,L)$ has n nodes, all of which are ordinary, and l links. The number of link flows that needs to be observed to achieve full observability is at least $(l-n+1)$.*

Proof: Strang (1986) noted that the rank of the node-link incidence matrix A of the traffic network is $(n-1)$. The dimensionality of the null space $N(A)$ of a matrix A of dimension, $n\times l$ is then

$$dim(N(A)) = l - r(A) = l-(n-1) = l-n+1 \tag{5}$$

This means that the solution to the homogeneous linear Equations (4) requires $(l-n+1)$ observations. That is to say, for a traffic network G, the number of links to be observed to achieve full observability is at least $(l-n+1)$.

Remark 1. The result of this theorem directly addresses the "Remarks and Open Problems" in Gu (2005). The original statement in Gu (2005) can be expressed as follows: "Let G be a planar digraph with l links and n nodes. It is proved that a minimum link control set of G contains $(l-n+1)$ links. Is this true for any digraph?" The conclusion of Theorem 1 provides the answer "yes, it is true for any digraph" regarding Gu's question.

The nodes in the traffic network cannot all be ordinary. When there are both centroid and ordinary nodes in the traffic network, the node flow conservation equation is written as the linear equations

$$A_{n\times l}V_{l\times 1} = s_n \rightarrow \begin{cases} a_{11}v_1 + a_{12}v_1 + \ldots + a_{1l}v_1 = s_1 \\ \vdots \\ a_{c1}v_1 + a_{c2}v_1 + \ldots + a_{cl}v_1 = s_c \\ a_{c+11}v_1 + a_{c+12}v_1 + \ldots + a_{c+1l}v_1 = 0 \\ \vdots \\ a_{n1}v_1 + a_{n2}v_1 + \ldots + a_{nl}v_1 = 0 \end{cases} \tag{6}$$

where the number of equations with expressions not equal to zero is the number of centroid nodes c, and the number of equations with expressions equal to zero is the number of ordinary nodes $(n-c)$.

Ng (2012) proposed transforming node incidence matrix A into the matrix A^* by removing the row associated with the centroid node. This transformation allows for reordering matrix A^* and solving linear equation group (5) similar to Eq. (4).

Ng (2012) proposed that the flow conservation equation for ordinary nodes be written in matrix form as

$$\begin{pmatrix} A^*_{no} & A^*_{ob} \end{pmatrix} \begin{pmatrix} V_{no} \\ V_{ob} \end{pmatrix} = 0 \tag{7}$$

where V_{no} is the set of link flows to be inferred for which no sensor is deployed; V_{ob} is the link flow set observed by the deployed sensors; A^*_{no} and A^*_{ob} are respectively submatrices of V_{no} and V_{ob} in the incidence matrix $A^*_{m\times l}$. Expanding Eq. (7) yields

$$A^*_{no}V_{no} + A^*_{ob}V_{ob} = 0 \tag{8}$$

If A^*_{no} is invertible, it follows from Eq. (8) that V_{no} can be obtained as

$$V_{no} = -{A^*_{no}}^{-1}A^*_{ob}V_{ob} \tag{9}$$

It should be noted that the critical problem is to ensure the existence of (and find) the invertible matrix A^*_{no}.

Corollary 1. *(Existence of A^*_{no}) There is an $(n-c)$-order invertible matrix A^*_{no} in the node-link incidence matrix A^* that deletes the centroid node.*

Remark 2. In Ng (2012), Corollary 1 is described as follows: "Proposition 1 (Existence of A^*_{no}). It is always possible to partition the matrix A^* into two matrices A^*_{no} and A^*_{ob}, where A^*_{no} is an n-by-n invertible matrix." A detailed proof has already been provided therein. Therefore, in this paper, we only restate this theorem using our symbols to show consistency with the result of Ng (2012).

Corollary 2. *It is assumed that traffic network $G=(N,L)$ has n nodes, with c centroid nodes and $(n-c)$ ordinary nodes. The rank of the ordinary node-link incidence matrix with deletion of the centroid node associated row is $(n-c)$.*

Theorem 2. *It is assumed that traffic network $G=(N,L)$ has n nodes, with c centroid nodes, $(n-c)$ ordinary nodes, and l links. The number of link flows to be observed for full observability is at least $(l+c-n)$.*

Proof: As in the proof of Theorem 1, it follows from the proof of Theorem 2 that the rank of the ordinary node-link incidence matrix of a traffic network $G = (N, L)$ with n nodes, of which $(n - c)$ are ordinary nodes, is $r(A^*) = n - c$. The dimensionality of the null space $N(A^*)$ of a matrix A^* of dimension $(n - c) \times l$ is then

$$dim(N(A^*)) = l - r(A^*) = l - (n - c) = l + c - n \tag{10}$$

This means that the solution of linear Eq. (4) requires $(l + c - n)$ observations; i.e., the number of link flows that needs to be observed to achieve full observability is at least $(l + c - n)$.

Remark 3. The conclusion in Theorem 2 is consistent with the result in Ng (2012): "Proposition 2 (Minimum number of sensors for full observability). Given a transportation network $G = (N, L)$. To observe all link flows in G, at a minimum, sensors need to be deployed on $(l - m)/l$*100%of the links." In this paper, we express Proposition 2 focusing solely on the minimum number of sensors required for full observability of the entire network, without considering the installation ratio. Additionally, we provide full proof whereas Ng (2012) omitted it.

Remark 4. According to Theorem 2, if the number of centroid nodes is high, then the number of links that needs to be observed to achieve full observability will also be high. That is to say, the number of links to be observed increases with the number of centroid nodes in the traffic flow observability problem under consideration.

4. Proposed algorithm for obtaining a virtual network

When there are centroid nodes and ordinary nodes in the traffic network, a lower bound for the number of sensors required to fully observe the flow of the entire traffic network can be determined. This lower bound is independent of the network topology and depends solely on the number of nodes and links within the traffic network. The following provides a minimum requirement for sensor deployment that enables full observation of network link flows, and it is hoped that this lower bound is associated with the network topology.

4.1. Defining a virtual network refactoring network

Through Corollary 1, and Theorem 2, the rank of the node-link incidence matrix of a traffic network can be calculated if all nodes in this network are ordinary nodes. When the centroid nodes and ordinary nodes simultaneously exist in one traffic network, it is impossible to fully consider all links in the full observability problem in the traffic network. Under this condition, it is necessary to delete the rows associated with the centroid nodes in the node-link incidence matrix. To make full use of the conservation of flow at the nodes and the consistency of traffic demand (i.e., the total travel yield and total travel attraction at the centroid), the method proposed by He (2103) is adopted to reconstruct the traffic network by adding a virtual centroid to replace many original centers of mass, and the method proposed by Castillo et al. (2014) is adopted to construct a non-plane hole generated network without centroid. However, He (2103) and Castillo et al. (2014) did not give a specific algorithm flow for a virtual network. We thus present the method of constructing a virtual network with a given network association.

First, a virtual node is added to the traffic network to delete the original centroid node and the link connecting the centroid node and the ordinary node. Without changing the flow direction and number of connecting links, a virtual link is added to connect the virtual node and ordinary nodes. The new traffic network constructed from virtual nodes and virtual links is called a virtual network.

According to the method of constructing the virtual network, compared with the original network, the number of links in the network does not change, and the direction of flow of the virtual links does not change from the original ordinary nodes. A small traffic network is depicted in Fig. 5a, which its associated virtual network is shown in Fig. 5b. Node 9 is a newly generated virtual node; the virtual links (4,9), (5,9), (5,9), and (6,9) respectively correspond to the links where traffic flows out from ordinary nodes 4, 5, 6 and flows into centroid nodes 7, 8, converted by links (4,7), (5,7), (5,8), and (6,8). The upper virtual link (9,9) and the lower virtual link (9,9) correspond to the link between centroid nodes 1 and 7, and the loop at centroid node 8, converted by (1,7) and (8,8) respectively. The virtual links (9,2) and (9,3) correspond to the links where traffic flows out from centroid node 1 and flows into ordinary nodes 2

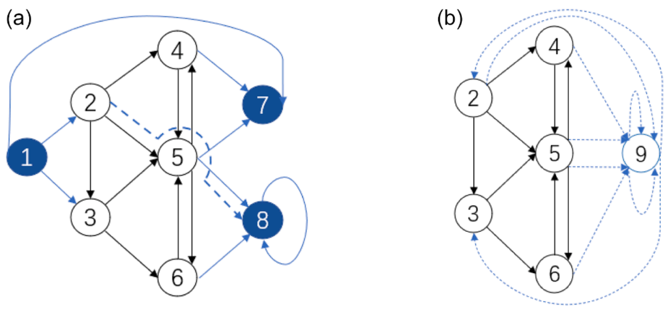


Fig. 5. (a). Planar graph of a small traffic network. (b). Associated virtual network.

and 3, converted by links (1,2) and (1,3) respectively.

The added virtual node is an ordinary node that satisfies the node flow conservation relationship because the previous virtual link connects the virtual centroid with the original ordinary node, and the total inflow to the virtual node is equal to the total outflow flow, which satisfies the flow conservation equation. For a traffic network $G = (N, L)$ with n nodes, c centroid nodes, $(n - c)$ ordinary nodes, and l links, the associated virtual network $\widetilde{G} = (\widetilde{N}, \widetilde{L})$ has $(n - c + 1)$ ordinary nodes and l links.

4.2. Virtual network computing method

According to the relevant knowledge of graph theory, the incidence matrix guarantees connections between nodes and links. The graph corresponding to the incidence matrix is isomorphic (unique). Therefore, the construction of virtual networks for large networks can be described by a node-link incidence matrix.

Step 1: Import the original graph to generate the node incidence matrix $A_{n \times l}$ of the original graph. A row of the matrix represents a node, and write $u_i, u_i \in N$. A column represents a link, and write $(u_i, u_j), (u_i, u_j) \in L$. A link (u_i, u_j) indicates that the direction of the link node u_i points to nodeu_j. The first $(n - c)$ rows of the matrix are the rows of ordinary nodes, and the last c rows are the rows of the centroid nodes.

$$A_{n\times l} = \begin{array}{l} \left.\begin{bmatrix} a_{1,1} & \cdots & a_{1,l} \\ \vdots & \ddots & \vdots \\ a_{n-c,1} & \cdots & a_{n-c,l} \end{bmatrix}\right\} \textit{rows of ordinary nodes} \\ \left.\begin{bmatrix} a_{n-c+1,1} & \cdots & a_{n-c+1,l} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,l} \end{bmatrix}\right\} \textit{rows of centroid nodes} \end{array} \tag{11}$$

Step 2: Generate the incidence matrix $\widetilde{A}_{(n-c+1)\ \times l}$ of the virtual network.
Step 2.1: The centroid node of the original incidence matrix A is changed into a virtual node without changing the node order. That is to say, if the nodes $u_i, u_j \in N \backslash N^*$, are ordinary nodes, $u_q \in N^*$ is a centroid node. $\widetilde{u}_q$ is an added virtual node, and the column index $(u_i, u_q), (u_q, u_j)$ becomes $(\widetilde{u}_q, u_q), (u_q, \widetilde{u}_q)$.
Step 2.2: The incidence matrix adds the row associated with the virtual node $\widetilde{u}_q$, and the associated row elements of the ordinary nodes do not change. The value obtained by adding each column element $\begin{bmatrix} a_{n-c+1,1} \\ \vdots \\ a_{n,1} \end{bmatrix}, \cdots, \begin{bmatrix} a_{n-c+1,l} \\ \vdots \\ a_{n,l} \end{bmatrix}$ of the rows in which the centroid nodes are located is placed in the virtual node row; i.e., the row element associated with the virtual node

$$\begin{bmatrix} \underbrace{a_{n-c+1,1} + \ldots + a_{n,1}}_{\textit{the number of centriod nodes } c} & \cdots & \underbrace{a_{n-c+1,l} + \ldots + a_{n,l}}_{\textit{the number of centriod nodes } c} \end{bmatrix} \tag{12}$$

The centroid rows are deleted in generating the node-link incidence matrix of the virtual network:

$$\widetilde{A}_{(n-c+1)\ \times l} = \begin{array}{l} \left.\begin{bmatrix} a_{1,1} & \cdots & a_{1,l} \\ \vdots & \ddots & \vdots \\ a_{n-c,1} & \cdots & a_{n-c,l} \end{bmatrix}\right\} \textit{rows of ordinary nodes} \\ \left.\begin{bmatrix} \underbrace{a_{n-c+1,1} + \ldots + a_{n,1}}_{\textit{the number of centriod nodes } c} & \cdots & \underbrace{a_{n-c+1,l} + \ldots + a_{n,l}}_{\textit{the number of centriod nodes } c} \end{bmatrix}\right\} \textit{rows of virtual nodes} \end{array} \tag{13}$$

Step 3: Using the Gaussian elimination algorithm by Hu (2009) and Ng (2012) to obtain a reduced row echelon form of the incidence matrix $\widetilde{A}$ (i.e., $rref(\widetilde{A})$), the observed link of the traffic network that requires the deployment of the sensor is solved.
Step 3.1: The Gaussian elimination algorithm (Anton, 1984) is used to obtain the reduced row echelon form of the matrix $\widetilde{A}$ (i.e., $rref(\widetilde{A})$) of the virtual network node-link incidence matrix $\widetilde{A}$

Table 1
Node-link incidence matrix A of the small network.

Link Node	1,2	1,3	2,3	2,4	2,5	3,5	3,6	4,5	4,7	5,7	5,8	6,5	6,8
2	1	0	−1	−1	−1	0	0	0	0	0	0	0	0
3	0	1	1	0	0	−1	−1	0	0	0	0	0	0
4	0	0	0	1	0	0	0	−1	−1	0	0	0	0
5	0	0	0	0	1	1	0	1	0	−1	−1	1	0
6	0	0	0	0	0	0	1	0	0	0	0	−1	−1
1	−1	−1	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0	1	1	0	0	0
8	0	0	0	0	0	0	0	0	0	0	1	0	1

Step 3.2: According to Eqs. (7), (8), and (9), Theorem 2 shows that there is an $(n-c)$-order invertible matrix in the node–link incidence matrix. Find the first entry of each row in the matrix $rref(\widetilde{A})$ that is not 0. There are $(n-c)$ columns, which can form a reversible matrix $\widetilde{A}_{no}$, and the associated links are marked as unobserved links.
Step 3.3: According to Eq. (9), the remaining links are the observed links that require the deployment of sensors in the traffic network.
Step 4: Using the incidence matrix mapping method, the virtual network graph is generated, and the links to be observed are marked according to the reduced row echelon form of the matrix. For a given incidence matrix $A_{n\times l}$, the method of drawing the corresponding graph is as follows.
Step 4.1: Make n isolated nodes.
Step 4.2: For all $1\le e\le l$, there are only two elements in $\{L_{1e}, L_{2e}, \ldots, L_{ne}\}$ that are non-zero values. If $a_{u_i,e}=-1, a_{u_j,e}=1$, and $u_i\ne u_j$, the directed link (u_i,u_j) connects nodes u_i and u_j, and the directed link (u_i,u_j) connects the nodes from the node u_i to node u_j.

A small traffic network shown in Fig. 3a is used as an example of the virtual network transformation. The network consists of 8 nodes and 13 links, where nodes 1, 7, and 8 are centroid nodes and the remaining nodes are ordinary nodes. The link (i,j) indicates that node i points to node j.

According to the steps of the virtual network computing method, the calculation result of each step for the concerned example is given.

Results in Step 1: Generate the node-link incidence matrix A according to Fig. 3a. Table 1 shows the association matrix A for a small network.

Centroid nodes 1, 7, and 8 are transformed into virtual node 9, and the associated links of nodes 1, 7, and 8, namely (1,2), (1,3), (4,7), (5,7), (5,8), and (6,8), are converted into virtual links (9,2), (9,3), (4,9), (5,9), (5,9), and (6,9). The associated rows with centroid nodes 1, 7, and 8 are deleted, and the elements in each column of these three rows are summed to generate the associated row for virtual node 9.

Results in Step 2: Obtain the node-link incidence matrix $\widetilde{A}$ of the virtual network by adding each column element of the three rows associated with the centroid node, shown in Table 2.

The italicized elements constitute the newly generated virtual node association row. The reduced row echelon form of the matrix $\widetilde{A}$ is obtained through Gaussian elimination.

Table 2
Node-link incidence matrix $\widetilde{A}$ of an associated virtual network of the small network.

Virtual link (Link) Node	9,2 (1,2)	9,3 (1,3)	2,3	2,4	2,5	3,5	3,6	4,5	4,9 (4,7)	5,9 (5,7)	5,9 (5,8)	6,5	6,9 (6,8)
2	1	0	−1	−1	−1	0	0	0	0	0	0	0	0
3	0	1	1	0	0	−1	−1	0	0	0	0	0	0
4	0	0	0	1	0	0	0	−1	−1	0	0	0	0
5	0	0	0	0	1	1	0	1	0	−1	−1	1	0
6	0	0	0	0	0	0	1	0	0	0	0	−1	−1
9	*−1*	*−1*	*0*	*0*	*0*	*0*	*0*	*0*	*1*	*1*	*1*	*0*	*1*

Table 3
The reduced row echelon form of the matrix $\widetilde{A}$ (i.e., $rref(\widetilde{A})$) of the associated virtual network of the small network.

Virtual link (Link) Node	9,2 (1,2)	9,3 (1,3)	2,3	2,4	2,5	3,5	3,6	4,5	4,9 (4,7)	5,9 (5,7)	5,9 (5,8)	6,5	6,9 (6,8)
2	**1**	**0**	−1	**0**	**0**	1	**0**	0	−1	−1	−1	1	0
3	**0**	**1**	1	**0**	**0**	−1	**0**	0	0	0	0	−1	−1
4	**0**	**0**	0	**1**	**0**	0	**0**	−1	−1	0	0	0	0
5	**0**	**0**	0	**0**	**1**	1	**0**	1	0	−1	−1	1	0
6	**0**	**0**	0	**0**	**0**	0	**1**	0	0	0	0	−1	−1
9	0	0	0	0	0	0	0	0	0	0	0	0	0

Results in Step 3: Calculate $rref(\widetilde{A})$, mark the first non-zero column in each row, and find the invertible matrix $\widetilde{A}_{no}$, shown in Table 3.

The bolded columns form an invertible matrix $\widetilde{A}_{no}$, and the corresponding links are unobserved links. Non-bolded columns represent observed links that require sensor deployment. It is necessary to deploy sensors on links (2,3), (3,5), (4,5), (4,7), (5,7), (5,8), (6,5), and (6,8).

Results in Step 4: Generate a virtual network, as shown in Fig. 6.

5. Proposed algorithm for obtaining a virtual network

The traffic network consists of two types of nodes: centroid and ordinary nodes. The centroids represent the origin and destination nodes of travel, introducing demand flow into the network, while also satisfying the law of conservation. However, the flow conservation equations at centroid nodes form a system of non-homogeneous linear equations, requiring consideration of the impact of OD demand when seeking a unique solution. This section initially investigates the impact of centroid nodes on virtual network construction before leveraging the connection relationships between nodes and links to establish a minimum requirement for sensor deployment, ensuring full observability of link flows.

5.1. Method of fully observing link flow based on the number of traffic network planes

- Planar network link flow is fully observable

Theorem 3. *It is assumed that planar traffic network $G = (N, L)$ has n nodes, with c centroid nodes, $(n - c)$ ordinary nodes, and l links. Construct a virtual network $\widetilde{G} = (\widetilde{N}, \widetilde{L})$ associated with the planar network G. The virtual network $\widetilde{G}$ has $(n - c + 1)$ ordinary nodes and l links. Assuming that the virtual network $\widetilde{G}$ has h finite planes, the number of link flows that needs to be observed to achieve full observability is at least h.*

Proof: According to the virtual network transformation steps, the virtual network $\widetilde{G}$ has $(n - c + 1)$ nodes and l links. These $(n - c + 1)$ nodes are ordinary nodes that satisfy the flow conservation that the total flow of inflow nodes equals the total flow of outflow nodes. According to Strang (1986), the rank of the node–link incidence matrix $\widetilde{A}$ of the virtual network $\widetilde{G}$ is

$$r(\widetilde{A}) = (n - c + 1) - 1 = n - c \tag{14}$$

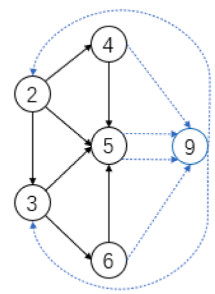


Fig. 6. Associated virtual network in this example.

According to the Euler's formula (Bondy, 1976), the relationship between the nodes, links, and number of planes in the virtual network $\widetilde{G}$ is

$$(n - c + 1) - l + h_0 = 2 \tag{15}$$

where h_0 are the sum of the numbers of finite planes and infinite planes in the virtual network; i.e., $h_0 = h + 1$.

Therefore, the relationship between the numbers of nodes, links, and finite planes in a virtual network $\widetilde{G}$ is

$$(n - c + 1) - l + (h + 1) = 2 \tag{16}$$

We introduce $h = l - n + c$; i.e., the number of finite planes is the number of links minus the number of all nodes plus the number of centroid nodes.

Therefore, the rank of the node-link incidence matrix $\widetilde{A}$ of the virtual network $\widetilde{G}$ is expressed as

$$r(\widetilde{A}) = n - c = l - h \tag{17}$$

The dimensionality of the null space $N(\widetilde{A})$ of a matrix $\widetilde{A}$ with dimensions $(n - c + 1) \times l$ is

$$dim(N(\widetilde{A})) = l - r(\widetilde{A}) = l - (l - h) = h \tag{18}$$

This means that the solution of linear Eq. (5) requires h observations; i.e., the number of link flows to be observed for full observability is at least h.

Corollary 3. *The number of virtual network $\widetilde{G}$ planes associated with the planar network G is equal to the number of linearly independent link flows in the network, which is the number of links plus the number of centroid nodes minus the number of all nodes; i.e., $h = l - n + c$.*

Proof: It is seen in the proof of Theorem 1 that for a planar network G with c centroid nodes, $(n - c)$ ordinary nodes, and l links, the rank of the ordinary node-link incidence matrix is $r(A^*) = n - c$, and the dimensionality of the null space $N(A^*)$ of matrix A^* having dimensions $(n - c) \times l$ is $dim(N(A^*)) = l - r(A^*) = l - (n - c) = l - n + c$. Therefore, the number of linear independent link flows in the planar network G is $(l - n + c)$.

It is seen in the proof of Theorem 3 that the virtual network G associated with the planar network G has $(n - c + 1)$ ordinary nodes and l links, and the rank of the node–link incidence matrix $\widetilde{A}$ of the virtual network $\widetilde{G}$ is $r(\widetilde{A}) = n - c = l - h$. The relationship between the numbers of nodes, links, and finite planes in the virtual network $\widetilde{G}$ is $n - c - l + h = 0$, and $h = l - n + c$ can be derived. Therefore, the number of planes in the virtual network is the number of links plus the number of centroid nodes minus the number of all nodes.

Therefore, the number of virtual networks $\widetilde{G}$ planes associated with the planar network G is equal to the number of linearly independent link flows in the network, which is the number of links plus the number of centroid nodes minus the number of all nodes; i.e., $h = l - n + c$.

Theorem 4. *It is assumed that planar traffic network $G = (N, L)$ has n nodes, with c centroid nodes, $(n - c)$ ordinary nodes, and l links. Assuming that the virtual network $\widetilde{G}$ has h finite planes, the number of link flows that needs to be observed to achieve full observability is at least h or $(l - n + c)$.*

Proof: The proof is obtained from Theorem 2 and Corollary 3.

Corollary 4. *The rank of the node-link incidence matrix $\widetilde{A}$ of the virtual network $\widetilde{G} = (\widetilde{N}, \widetilde{L})$ is given by the number of finite planes in the network; i.e., $r(\widetilde{A}) = l - h$.*

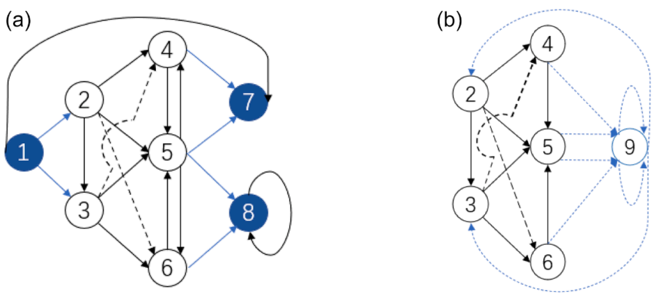


Fig. 7. (a) Non-planar graph of a small traffic network. (b) Associated virtual network of a non-planar network.

Proof: The proof is obtained from Theorem 3.

- Non-planar network link flow is fully observable

When the connection link of the non-planar graph is connected to a centroid node and ordinary nodes, the graph is changed into a virtual network to become a planar graph. When the connecting link of the non-planar graph appears as two ordinary nodes are connected, the virtual network is still a non-planar graph, and thus it is still necessary to consider that the link flow of the non-planar graph is fully observable.

It can be considered that a non-planar network is constructed by adding intersecting links to the planar network. For non-planar networks, the number of sensors that needs to be deployed to fully observe the flow of planar links can be calculated first, and the number of links that needs to be added from the plane to the non-planar network is then added.

(1) When a non-planar network adds a link connecting a centroid node to an ordinary node (as shown in Fig. 5a, where the added link (2,8) connects ordinary node 2 with centroid node 8), the transformation to a virtual network can convert it into a planar network (as shown in Fig. 5b). The number of observed link flows required to achieve full observability of link flows can be calculated according to a planar network.
(2) When a non-planar network adds two links that connect only ordinary nodes (as shown in Fig. 7, where the added links (2,6) and (3,4) connect ordinary nodes 2 with 6, and 3 with 4), the transformed virtual network remains non-planar (as shown in Fig. 7b). The effect of adding links to the links that need to be observed to achieve full observability of the link flow needs to be considered.

Theorem 5. *It is assumed that a non-planar network $\widehat{G} = (\widehat{N}, \widehat{L})$ has n nodes with c centroid nodes, $(n - c)$ ordinary nodes, and l links. When the transformed virtual network is planar, has h planes. The number of link flows that needs to be observed to achieve full observability is at least h or $(l - n + c)$. When the transformed virtual network is still non-planar, has l links with k links are added links and h planes. The number of link flows that needs to be observed to achieve full observability is at least $(h + k)$ or $(l - n + c)$.*

Proof: According to Strang (1986), the non-planar network $\widehat{G} = (\widehat{N}, \widehat{L})$ has n nodes and l links, and the rank of its node–link incidence matrix $\widehat{A}$ is $r(\widehat{A}) = n - 1$. Corollary 2 shows that if there are c centroid nodes and $(n - c)$ ordinary nodes among n nodes, the rank of the ordinary node–link incidence matrix $\widehat{A}^*$ is $r(\widehat{A}^*) = n - c$. Theorem 3 shows that the non-planar network $\widehat{G} = (\widehat{N}, \widehat{L})$ has n nodes, with c being centroid nodes and $(n - c)$ being ordinary nodes, and l links. The number of link flows that needs to be observed to achieve full observability of the link flow is at least $(l - n + c)$.

Corollary 3 shows that the number of virtual networks $\widehat{G}$ planes associated with a planar network G is equal to the number of linearly independent link flows in the network, which is the number of links minus the number of ordinary nodes; i.e., $h = l - n + c$. According to Theorem 3, the non-planar network $\widehat{G} = (\widehat{N}, \widehat{L})$ has n nodes, with c being centroid nodes $(n - c)$ being ordinary nodes, and $(l + k)$ links.

(1) When the transformed virtual network is planar, according to the planar network calculation method, the number of link flows that needs to be observed to achieve full observability of the link flow is at least h or $(l - n + c)$.
(2) When the transformed virtual network is still non-planar, considering the addition of links, the number of links that needs to be observed to achieve full observability of the link flow is at least $(h + k)$ or $(l - n + c)$.

5.2. Method of fully observing link flow under special circumstances

- Full link flow observability with ordinary nodes

Theorem 1 shows that for a traffic network, the number of link flows to be observed for full observability is at least $(l - n + 1)$. Because all nodes are ordinary nodes, it is not necessary to carry out a virtual network transformation and only necessary to calculate the number of planes in the traffic network itself. Euler's formula shows that if the numbers of nodes, links, and planes in a connected planar graph G are n, l, and h_0 respectively, then $n - l + h_0 = 2$ and $h_0 = h + 1$, such that $h = l - n + 1$.

Theorem 6. *It is assumed that network $G = (N, L)$, has n nodes, all of which are ordinary nodes, and l links. When G is a planar traffic network, has h planes. The number of link flows that needs to be observed to achieve full observability is at least $(l - n + 1)$ or h; when G is a non-planar traffic network, has l links with k links are added links and h planes. The number of link flows that needs to be observed to achieve full observability is at least $(l - n + 1)$ or $(h + k)$.*

Proof: Euler's formula shows that $h = l - n + 1$. Theorem 1 shows that for the traffic network G, the number of link flows that needs to be observed to achieve full observability of the link flow is at least $(l - n + 1)$.

Therefore, for a planar traffic network G with n nodes, all of which are ordinary, and l links, the number of links that needs to be observed to achieve full observability of the link flow is at least $(l - n + 1)$ or h.

When the graph is a non-planar network G with k added links, according to Theorem 4, the number of links that needs to be observed to achieve full observability of the link flow is at least $(l - n + 1)$ or $(h + k)$.

- Full link flow observability with centroid nodes

Achieving a unique solution for the flow conservation equations at centroid nodes requires taking into account the influence of OD demand. However, in networks with a high volume of centroid nodes, or when every node is designated as a centroid, the number of homogeneous linear equations resulting from node flow conservation can be significantly reduced, or may even reach zero.

If all nodes in the traffic network are centroids, the node flow conservation equation is written as a linear system of equations:

$$A_{n\times l}V_{l\times 1} = s_n \rightarrow \begin{cases} a_{11}v_1 + a_{12}v_1 + \ldots + a_{1l}v_1 = s_1 \\ \vdots \\ a_{c1}v_1 + a_{c2}v_1 + \ldots + a_{cl}v_1 = s_n \end{cases} \tag{19}$$

The equations are non-homogeneous linear equations, and the number of equations is equal to the number of unknowns, which is equal to the number of centroid nodes n. If the non-homogeneous linear equations are to be solved, the sum of the equilibrium flow of all centroids must be zero. From linear algebra, the unique solution of the nonhomogeneous linear equations is

$$r(A_{n\times l}) = r(A_{n\times l}, s_n) = n \tag{20}$$

When we also regard the traffic network with all nodes as the centroid as a directed connected graph, the rank of the node-link incidence matrix remains $(n - 1)$, which is contradictory to the case that there is a unique solution.

According to Corollary 1, the incidence matrix A^* is of order $(n - c) \times l$, with $l > (n - c)$ generally, and the invertible matrix A^*_{no} should be of order $(n - c) \times (n - c)$; i.e., an ordinary node corresponds to a link that is not deployed with sensors.

Thus, it is postulated that when both centroid and ordinary nodes coexist within a traffic network, the number of links that must be inferred in the absence of sensors corresponds to the count of ordinary nodes. Conversely, if every node within the traffic network is a centroid node, there are no ordinary nodes present, implying that the number of links requiring sensor-less inference also becomes zero. In such a scenario, surveillance through sensors is essential for all links within the traffic network.

Theorem 7. *It is assumed that network $G = (N, L)$ has n nodes, all of which are centroid nodes, and l links. The number of link flows that needs to be observed to achieve full observability is at least l.*

Proof: All nodes in the traffic network are centroid nodes; i.e., $n = c$. According to Theorem 2, the number of link flows that needs to be observed in the traffic network to achieve full observability $(l - n + c)$ is such that $l - n + c = l$.

Given a traffic network where all nodes are centroids, the number of links that needs to be observed to achieve full observability is the number of all links in the network.

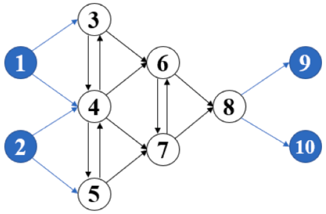
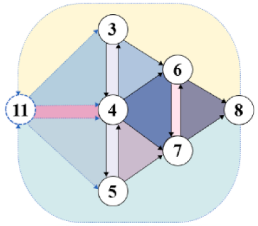
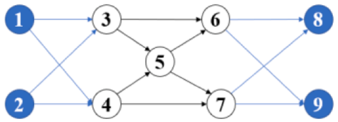
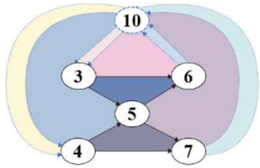
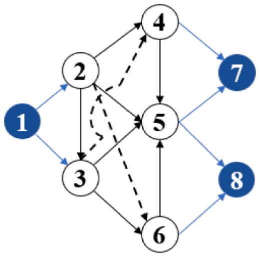
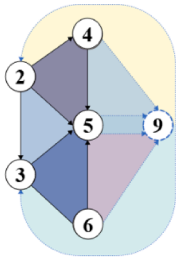
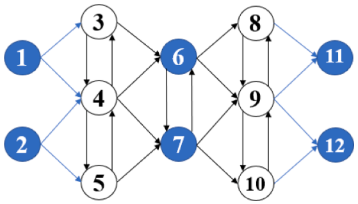
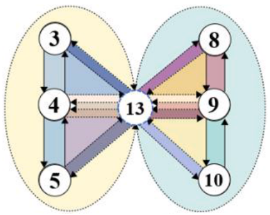
As shown in Table 4, there are differences between this study and the research by Castillo et al. (2014).

Table 5 shows several traffic networks; some were used by Hu et al. (2009) and some were modified. Columns 2 through 7 show the associated planar virtual network, links, nodes, several added links, and several planes in the virtual networks. Finally, column 8 gives the number of link flows to be observed so that the links are fully observable. We note that the same results were obtained when other researchers' formulas were applied.

Table 4
The differences between this study and the research by E. Castillo et al. (2014).

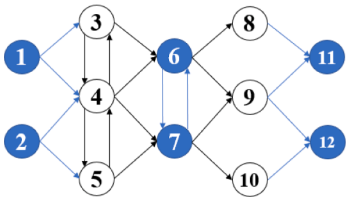
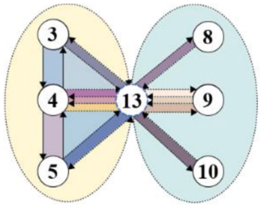
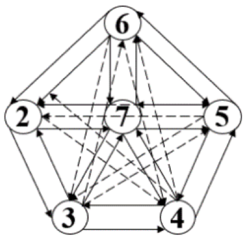
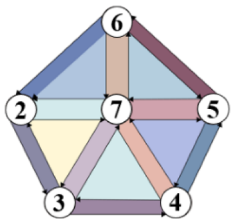
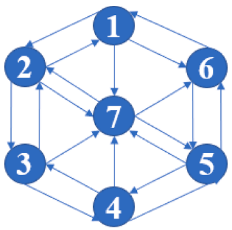
	E. Castillo et al. (2014)	This study
Planar network	$\dot{h} + c - 1$ $l - n + c$	h $l - n + 1; l - n + c$
Non-planar network	$\dot{h} + k + c - 1$ $l - n + c$	$h + k$ $l - n + 1; l - n + c$
Difference between hole $\dot{h}$ and plane h	$\dot{h}$ represents the internal plane of the traffic network itself.	h represents the inner plane of a virtual network associated with a traffic network.

Table 5
Links, nodes, centroid nodes, added links, virtual network finite planes, and number of links to be counted for full link observability in various traffic networks.

Network	Planar virtual network	Links l	Nodes n	Centroid nodes c	Added links k	Virtual network finite planes h	observed $h + k$/ l-$n + c$
		18	10	4	0	12	12
		12	9	4	0	9	9
		15	8	3	2	8	10
		26	12	6	0	20	20

(*continued on next page*)

Table 5 (*continued*)

Network	Planar virtual network	Links l	Nodes n	Centroid nodes c	Added links k	Virtual network finite planes h	observed $h + k/$ $l\text{-}n + c$
		22	12	6	0	16	16
		30	6	0	10	15	25
		20	7	7	0	0	20

6. Graph theory and matrix analysis node flow conservation

Starting from the basic concepts of graph theory, we can understand the connectivity and structure of traffic networks. By defining a matrix to describe the node flow conservation equation, the traffic network is represented as a matrix system, which provides a theoretical basis for the analysis of the traffic flows in the network.

6.1. Relevant definitions and concepts

Definition 6. **(Path and loop)** $G = (N, L)$ as an undirected graph. The alternating sequence of nodes and links in G, $\Gamma = u_{i_0} e_{j_1} u_{i_1} e_{j_2} \cdots e_{j_k} u_{i_k}$ called u_{i_0} to u_{i_k} path, where $u_{i_{r-1}}$ and u_{i_r} are end points of e_{j_r}, for $r = 1, 2, \cdots, k$. The nodes u_{i_0} and u_{i_k} are referred to as the begin and end nodes of Γ respectively, and the number of links of Γ is called its length. If $u_{i_0} = u_{i_k}$, the Γ is called a loop. A simple path refers to a path where all links are different. If a simple path has the same start and end node ($u_{i_0} = u_{i_k}$), it is called a simple loop. A primary path or path consists of different nodes (except when u_{i_0} and u_{i_k} maybe the same), with all links being distinct. When this primary path has the same start and end node ($u_{i_0} = u_{i_k}$), it is referred to as a primary loop or loop.

The loop can be simply understood as a closed path. In planar graphs, the defined "plane" is also a special loop, which does not contain other links. All loops of the same length are isomorphic, in the sense that there is only one loop of a given length.

Definition 7. **(Spanning subgraph)** $G = (N, L)$ and $G' = (N', L')$ for two directed graphs, if $N' \subseteq N$ and $L' \subseteq L$, called G' a subgraph of G, G is called the supergraph of G', denoted as $G' \subseteq G$. If $N' \subset N$ or $L' \subset L$, then G' is said to be the proper subgraph of G. If $N' = N$, then G' is said to be a spanning subgraph of G.

Similar to a subset, a subgraph is a part of a graph, formed by removing certain nodes and links from the graph. The spanning subgraph G' contains all nodes of the graph G, but not all links.

Definition 8. **(Cut-set)** A cut-set is a subset of links in a graph G that satisfies the following conditions: (1) After removing all links of the cut-set, the graph G is divided into two parts; (2) If any link in the cut-set is not removed, the graph remains connected. Graph theory states that after a link is removed, the nodes at both ends cannot be removed.

Definition 9. **(Spanning tree)** A subgraph T of a graph G is called a spanning tree if it satisfies the following conditions: (1) T is connected; (2) T includes all nodes of the graph; (3) T has no loops. The links of G in T are called the branch of T, and the links not in T are called the chord of T.

Proposition 2. *For a connected graph $G = (N, L)$ with n nodes and l links, it can be proved that the number of branches of any spanning tree can be expressed as $t = n - 1$; the number of chords can be expressed as $x = l - (n - 1) = l - n + 1$.*

Proof: We prove by induction. When $n = 1$, the proposition is true. Suppose the proposition holds for $n \leq k (k \geq 1)$.

When $n = k + 1$, let $e_i = (u_i, u_j)$ be a link in T. Since there are no loops in T, removing e_i divides $T - e_i$ into two connected components T_1 and T_2.

Let n_i and t_i denote the number of nodes and links in T_i, respectively, then $n_i \leq k$. By the induction hypothesis, $t_i = n_i - 1$, for $i = 1, 2$. Thus, $t = t_1 + t_2 + 1 = n_1 + n_2 - 2 + 1 = n - 1$.

Therefore, the proposition also holds when $n = k + 1$. the number of chords $x = l - (n - 1) = l - n + 1$.

Remark: Given proposition 2, the number of branches of the spanning tree is $t = n - 1$, which is consistent with the result of proposition 6 in He (2013). This proposition provides a new way of using the spanning tree to prove He's result.

Definition 10. **(Fundamental cut-set system)** Let G be a graph of order n with a spanning tree T. Let $e_1, e_2, \cdots, e_{n-1}$ be the branches of T, and $C_i (i = 1, 2, \cdots, n - 1)$ be the cut-set formed by the branch e_i and the chords. Then C_i is called the fundamental cut-set corresponding to the branch e_i of G. The set $\{C_1, C_2, \cdots, C_{n-1}\}$ forms the fundamental cut-set system corresponding to T in G.

Definition 11. **(Fundamental loop system)** Let T be a spanning tree of a graph G with n nodes and l links. Let $e'_1, e'_2, \cdots, e'_{l-n+1}$ be the chords of T. Let $M_r (r = 1, 2, \cdots, l - n + 1)$ be the loop formed in G by adding the chord e'_r to T, consisting of the chord e'_r and the branches. M_r is called the fundamental loop corresponding to the chord e'_r of G. The set $\{M_1, M_2, \cdots, M_{l-n+1}\}$ forms the fundamental loop system corresponding to T in G.

The number of nodes n of graph G is its rank, the number of branches t is its cut-set rank, and the number of chords x is its loop rank. The fundamental loop consists of branches plus a chord, each chord corresponds to only one fundamental loop. The fundamental cut-set consists of several (not necessarily all) chords plus a branch, each branch corresponds to only one fundamental cut-set. The fundamental cut-set and the fundamental loop are independent of each other.

6.2. The relationship between cut-set matrix, loop matrix, plane matrix, and incidence matrix

In a directed graph, links are associated with nodes, loops, and cut-sets, each with distinct relationships that can be represented using different matrices. By defining the fundamental loop matrix, the fundamental cut-set matrix, and the plane matrix, and interconnecting them using the incidence matrix, it is possible to comprehensively describe and analyze the flow conservation properties within traffic networks.

The incidence matrix A represents the relationship between nodes and links, where rows represent nodes and columns represent links.

$$a_{ue} = \begin{cases} 1 & e \in O_u \\ -1 & e \in D_u \\ 0 & otherwise \end{cases} \tag{21}$$

According to Strang (1986), the sum of each row of matrix A is a zero row, indicating that the rows of A are not linearly independent. Hence, the rank of the incidence matrix A is $(n-1)$. Therefore, removing any row from A results in a $(n-1)\times l$ matrix $A^{'}$, which is called the reduced incidence matrix.

Definition 12. **(Fundamental incidence matrix)** Let $G=(N,L)$ be a directed connected graph with n nodes and l links. Deleting the row corresponding to the node u_i from the incidence matrix A, results in a reduced incidence matrix $A^{'}$. The columns of the matrix $A^{'}$ are then rearranged, placing the chords columns in the front and the branches columns at the back, the final association matrix $\mathscr{A}$ written by the spanning tree is obtained, which is called $\mathscr{A}=\{a_{u_ie_j}\}_{(n-1)\ \times l}$ is a fundamental incidence matrix.

$$a_{u_ie_j} = \begin{cases} 1 & \text{the link } e_j \text{ is connected to the node } u_i \text{ and departs from } u_i \\ -1 & \text{the link } e_j \text{ is connected to the node } u_i \text{ and points to } u_i \\ 0 & \text{the link } e_j \text{ is not connected to the node } u_i \end{cases}$$

Given graph G, a fundamental incidence matrix $\mathscr{A}$ can be uniquely determined, and conversely, providing matrix $\mathscr{A}$ allows for the unique construction of graph G (or its isomorphic graph).

$$\mathscr{A} = A_x A_t \tag{22}$$

Where A_t represents the association between branches and nodes, and A_x represents the association between chords and nodes. Since the number of branches is $t=n-1$, and the number of chords is $x=l-n+1$, A_t is an $(n-1)$-order square matrix, and A_x is an $(l-n+1)\times(n-1)$-order matrix.

Proposition 3. ***(The invertibility of matrix A_t)*** *In a connected graph $G=(N,L)$ with n nodes and l links, the fundamental incidence matrix $\mathscr{A}$ written from the spanning tree T with node u_i as the root is a row full rank, with a rank of $(n-1)$. Furthermore, the $(n-1)$-order submatrix A_t consisting of columns from branches is non-singular, and also has a rank of $(n-1)$.*

Proof: Through $|A_t|$ prove by induction. Assume $n=2$, i.e., graph G is a simple link, $|A_t|=\pm1$. Assume that for the graph with $(n-1)$ nodes, $|A_t|=\pm1$ holds.

To form a spanning tree with n nodes, a new node must be connected to only one node from the original spanning tree, or else a closed loop would form.

This insertion in A_t is represented by inserting a row and a column at the far right of the original A_t. The first $(n-1)$ elements in this row are 0 and the n^{th} element is ± 1.

Calculating the determinant, we find that for a spanning tree with n nodes, $|A_t|=\pm1$ still holds.

Thus, A_t is non-singular, with a rank of $(n-1)$, and consequently, the incidence matrix $\mathscr{A}$ is a row full rank with a rank of $(n-1)$.

Lemma 1. ***(Binet-Cauchy Theorem,*** *Brualdi, 1983) Let A be an $n\times s$ matrix and B be an $s\times n$ matrix. Then*

$$|AB| = \begin{cases} 0 & n > s \\ |A||B| & n = s \\ \sum\limits_{1\le k_1<k_2<\cdots<k_n\le s} |A(12\cdots n;k_1k_2\cdots k_n)||B(k_1k_2\cdots k_n;12\cdots n)| & n < s \end{cases}$$

where $|A(12\cdots n;\ k_1k_2\cdots k_n)|$ denotes the determinant of the submatrix of A formed by the columns indexed $k_1k_2\cdots k_n$; and $|B(k_1k_2\cdots k_n;\ 12\cdots n)|$ denotes the determinant of the submatrix of B formed by the rows indexed $k_1k_2\cdots k_n$.

Theorem 8. ***(Number of sensor deployment schemes)****: Let $\mathscr{A}$ be a fundamental incidence matrix of a directed connected graph $G=(N,L)$. Then, the number of different spanning trees in G is given by $|\mathscr{A}\mathscr{A}^T|$. Therefore, there are $|\mathscr{A}\mathscr{A}^T|$ possible sensor deployment schemes in the traffic network.*

Proof: Let $\mathscr{A}=(a_{ij})_{(n-1)\ \times l}$, $(n-1)\le l$, where n is the number of nodes and l is the number of links in the directed graph G. According to the Binet-Cauchy theorem (Lemma 1), the determinant of the product of $\mathscr{A}$ and its transpose $\mathscr{A}^T$, denoted as $|\mathscr{A}\mathscr{A}^T|$, can be expanded as a sum over all $(n-1)$-order submatrices of $\mathscr{A}$:

$$|\mathscr{A}\mathscr{A}^T| = \sum_i |\mathscr{A}_i||\mathscr{A}_i^T| \tag{23}$$

Here, $|\mathscr{A}_i|$ represents the determinant of an $(n-1)$-order submatrix of $\mathscr{A}$, and $|\mathscr{A}_i^T|$ represents the determinant of the corresponding

$(n-1)$-order submatrix of $\mathscr{A}^T$. Since $\mathscr{A}^T$ is the transpose of $\mathscr{A}$, the determinant of the submatrix in $\mathscr{A}^T$ corresponding to the j-th row is the same as the determinant of the submatrix in $\mathscr{A}$ corresponding to the j-th column, that is, $|\mathscr{A}_i| = |\mathscr{A}_i^T|$.Therefore, Eq. (23) can be rewritten as:

$$\left|\mathscr{A}\mathscr{A}^T\right| = \sum_i |\mathscr{A}_i|^2 \tag{24}$$

In the proof of Theorem 8, it is shown that $|A_t| = \pm 1$, so $|\mathscr{A}_i|^2 = 1$. This indicates that if the links corresponding to the columns of $\mathscr{A}_i$ form a spanning tree in G, then the contribution to $|\mathscr{A}\mathscr{A}^T|$ is 1. Eq. (24) sums over all combinations of $|\mathscr{A}_i|^2$, thus $|\mathscr{A}\mathscr{A}^T|$ represents the number of different spanning trees in G.

According to Definition 12 and Proposition 3, it can be seen that the matrix A_t is related to a particular sensor deployment scheme and different A_t corresponds to different sensor location schemes. Therefore, according to Eq. (24), the total number of possible sensor deployment schemes is equal to $|\mathscr{A}\mathscr{A}^T|$.

Definition 13. (Fundamental cut-set matrix) A matrix consisting of all cut-sets of a directed connected graph G is called the complete cut-set matrix, denoted as Q. Given a spanning tree T of the graph G, the matrix consisting of all fundamental cut-sets is called the fundamental cut-set matrix, denoted as $\mathscr{Q}$. The matrix $\mathscr{Q}$ represents the relationship between links and cut-sets, where each row corresponds to a fundamental cut-set and each column corresponds to a link. Each branch of the spanning tree corresponds to only one fundamental cut-set, thus the number of fundamental cut-sets is $t = n-1$, and therefore $\mathscr{Q} = \{q_{e_iC_i}\}_{(n-1)\times l}$

$$q_{e_iC_i} = \begin{cases} 1 & \text{The link } e_i \text{ is in the fundamental cut} - \text{set } C_i \text{ and has either the same direction} \\ -1 & \text{The link } e_i \text{ is in the fundamental cut} - \text{set } C_i \text{ and has either the opposite direction} \\ 0 & \text{The link } e_i \text{ is not in the fundamental cut} - \text{set } C_i \end{cases}$$

Since a fundamental cut-set is a single-branch cut-set, meaning each cut-set contains exactly one branch. In the matrix $\mathscr{Q}$, the columns corresponding to branches will have only one row with a value of 1 or -1, while the rest of the rows will be 0. Suppose when constructing the fundamental cut-set matrix, the columns are rearranged such that the chords' columns come first, followed by the branches' columns, and the branches' columns are aligned with the cut-sets, then the branch part will exactly form the identity matrix I_t, i.e.,

$$\mathscr{Q} = Q_f I_t \tag{25}$$

where Q_f represents the association between chords and nodes, the $(n-1)$-order identity matrix I_t represents a spanning tree. The rows of $\mathscr{Q}$ are full rank and the $rank(\mathscr{Q}) = n-1$.

Definition 14. (Fundamental loop matrix) A matrix consisting of all loops of a directed connected graph G is called the complete loop matrix, denoted as H. Given a spanning tree, T of the graph G, the matrix consisting of all fundamental loops is called the fundamental loop matrix, denoted as $\mathscr{H}$. The matrix $\mathscr{H}$ represents the relationship between links and loops, where each row corresponds to a fundamental loop, and each column corresponds to a link. A fundamental loop is formed by adding a chord to a branch; thus, the number of fundamental loops is $x = l-n+1$. Therefore, $\mathscr{H} = \{h_{e_rm_r}\}_{(l-n+1)\times l}$

$$h_{e_rm_r} = \begin{cases} 1 & \text{The link } e_r \text{ is in the fundamental loop } M_r \text{ and has either the same direction} \\ -1 & \text{The link } e_r \text{ is in the fundamental loop } M_r \text{ and has either the opposite direction} \\ 0 & \text{The link } e_r \text{ is not in the fundamental loop } M_r \end{cases}$$

Similar to the fundamental cut-set matrix, assuming that the direction of the loops is consistent with the direction of the chords, the branch columns exactly form an $(l-n+1)$-order identity matrix. After rearranging the columns, i.e.,

$$\mathscr{H} = I_x H_t \tag{26}$$

where I_x represents the association between chords and nodes as an $(l-n+1)$-order identity matrix, and H_t represents a spanning tree. The rows of $\mathscr{H}$ are full rank and $rank(\mathscr{H}) = l-n+1$.

Definition 15. (Plane Matrix) For a planar graph, it is observed that the finite planes defined in the graph are a special type of loop. From the complete loop matrix C, the rows corresponding to the loops forming finite planes are extracted to form an $(l-n+1)\times l$ order matrix $\mathscr{P}$. The matrix $\mathscr{P}$ represents the relationship between links and finite planes, where each row corresponds to a finite plane, and each column corresponds to a link. The number of finite planes is $h = l-n+1$, thus $\mathscr{P} = \left\{p_{e_jm_j}\right\}_{(l-n+1)\times l}$

$$p_{e_jm_j} = \begin{cases} 1 & \text{The link } e_j \text{ is in the plane } M_j \text{ and has either the same direction} \\ -1 & \text{The link } e_j \text{ is in the plane } M_j \text{ and has either the opposite direction} \\ 0 & \text{The link } e_j \text{ is not in the plane } M_j \end{cases}$$

Network finite planes are not necessarily fundamental loops, so there's no special column rearrangement to simplify them. However, the collection of all nodes and links in any two planes is no longer a plane of the network. Hence, each finite plane in the graph is independent of each other, meaning the rows of $\mathscr{P}$ are full rank, i.e., $rank(\mathscr{P}) = l - n + 1$.

Theorem 9. *For a traffic network $G = (N, L)$, has n nodes consisting of $(n - c)$ ordinary nodes and c centroid nodes, and l links. Its virtual network has $(n - c + 1)$ nodes, all of which are ordinary nodes, and l links. By removing rows related to centroid nodes and choosing a virtual node as the root, the spanning tree T at this virtual node consists of $(n - c + 1)$ nodes, $(t = n - c)$ branches, and $(x = l - n - c)$ chords. The fundamental incidence matrix $\mathscr{A} = \{a_{ue}\}_{(n-c)\ \times l}$, fundamental cut-set matrix $\mathscr{Q} = \{q_{e_iC_i}\}_{(n-c)\ \times l}$, fundamental loop matrix $\mathscr{H} = \{h_{e_rm_r}\}_{(l-n+c)\ \times l}$, and plane matrix $\mathscr{P} = \left\{p_{e_jm_j}\right\}_{(l-n+c)\ \times l}$ have the following relationship:*

$$\begin{cases} \mathscr{A}\mathscr{H}^{\mathrm{T}} = 0_{(n-c)\ \times\ (l-n+c)} \\ \mathscr{Q}\mathscr{H}^{\mathrm{T}} = 0_{(n-c)\ \times\ (l-n+c)} \\ \mathscr{A}\mathscr{P}^{\mathrm{T}} = 0_{(n-c)\ \times\ (l-n+c)} \\ \mathscr{Q}\mathscr{P}^{\mathrm{T}} = 0_{(n-c)\ \times\ (l-n+c)} \end{cases}$$

$$\begin{cases} \mathscr{H}\mathscr{A}^{\mathrm{T}} = 0_{(l-n+c)\ \times\ (n-c)} \\ \mathscr{H}\mathscr{Q}^{\mathrm{T}} = 0_{(l-n+c)\ \times\ (n-c)} \\ \mathscr{P}\mathscr{A}^{\mathrm{T}} = 0_{(l-n+c)\ \times\ (n-c)} \\ \mathscr{P}\mathscr{Q}^{\mathrm{T}} = 0_{(l-n+c)\ \times\ (n-c)} \end{cases}$$

The fundamental incidence matrix $\mathscr{A}$, fundamental cut-set matrix $\mathscr{Q}$, fundamental loop matrix $\mathscr{H}$, and plane matrix $\mathscr{P}$ reveal relationships that simplify the network's structure by matrix equations, making it easier to analyze traffic flow relationships.

Proof: Assuming $\mathscr{H}\mathscr{Q}^{\mathrm{T}} = B$, according to the rules of matrix multiplication, we can deduce that any element of B

$$b_{M_jC_r} = \sum_{e_i=1}^{l} h_{C_re_i} q'_{e_iM_j} \tag{27}$$

where $h_{C_re_i}$ is an element of the fundamental cut-set matrix $\mathscr{H}$, $q'_{e_iM_j}$ an element in the transpose $\mathscr{Q}^T$ of the fundamental loop matrix $\mathscr{Q}$, Obviously, $q'_{e_iM_j} = q_{M_je_i}$, $q_{M_je_i}$ is an element in $\mathscr{Q}$. Hence,

$$b_{M_jC_r} = \sum_{e_i=1}^{l} h_{C_re_i} q_{M_je_i} \tag{28}$$

Eq. (28) sums over all links, but it's apparent that it only needs to account for links within the cut-set C_r and loop M_j. Links belonging to both cut-set C_r and loop M_j must occur in even numbers as they appear in pairs; otherwise, loop M_j cannot close. Each pair of these links nullifies when their corresponding elements are multiplied, thus $b_{M_jC_r} = 0$, meaning $B = 0$. Therefore, this demonstrates that $\mathscr{H}\mathscr{Q}^T = 0$.

Utilizing graphical representations, consider the fundamental cut-set matrix $\mathscr{Q}$ and the fundamental loop matrix $\mathscr{H}$, where their columns are organized based on the link order. $\mathscr{Q}$ delineates the relationship between fundamental cut sets and links, while $\mathscr{H}$ illustrates the relationship between fundamental loops and links. When links e_i and e_j are concurrently part of the cut-set C_r and loop M_j:

(1) If both links e_i and e_j align with or oppose the direction of the cut-set C_r, both corresponding elements in $\mathscr{Q}$ are simultaneously 1 or -1. Conversely, if links e_i and e_j oppose each other's directions concerning loop M_j, one element in $\mathscr{H}$ is 1 while the other is -1;

In Fig. 8a, if $q_{M_je_i} = q_{M_je_j} = 1$, then $h_{C_re_i} = 1$ and $h_{C_re_j} = -1$. Conversely, if $q_{M_je_i} = q_{M_je_j} = -1$, then $h_{C_re_i} = -1$ and $h_{C_re_j} = 1$.

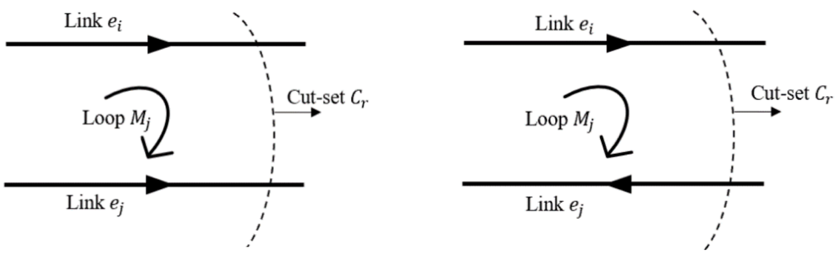


Fig. 8a. Correspondence between link directions and elements in $\mathscr{Q}$ and $\mathscr{H}$ matrices.

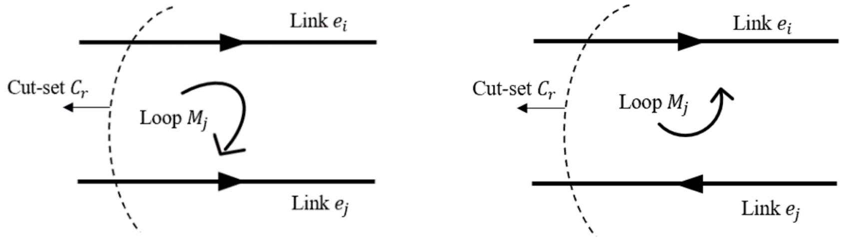


Fig. 8b. Correspondence between link directions and elements in $\mathscr{Q}$ and $\mathscr{H}$ Matrices.

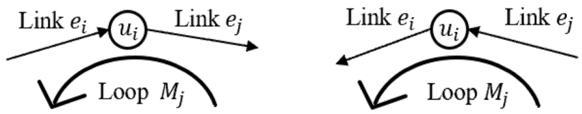


Fig. 9a. Correspondence between link directions and elements in $\mathscr{A}$ and $\mathscr{H}$ Matrices.

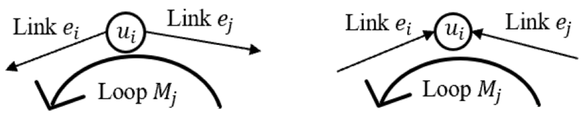


Fig. 9b. Correspondence between link directions and elements in $\mathscr{A}$ and $\mathscr{H}$ Matrices.

(2) If both links e_i and e_j align with or oppose the direction of the loop M_j, both corresponding elements in $\mathscr{H}$ are simultaneously 1 or -1. Conversely, if links link e_i and e_j oppose each other's directions concerning cut-set C_r, one element in $\mathscr{Q}$ is 1 while the other is -1.

In Fig. 8b, if $h_{C_r e_i} = h_{C_r e_j} = 1$, then $q_{M_j e_i} = 1$ and $q_{M_j e_j} = -1$. Conversely, if $h_{C_r e_i} = h_{C_r e_j} = -1$, then $q_{M_j e_i} = -1$ and $q_{M_j e_j} = 1$. That is, make the product of the corresponding row vectors of matrices $\mathscr{Q}$ and $\mathscr{H}$ equal to zero, which is represented as $\mathscr{H}\mathscr{Q}^T = 0_{(n-c) \times (l-n+c)}$.

$$\mathscr{Q}\mathscr{H}^T = \left[\mathscr{H}\mathscr{Q}^T\right]^T = 0_{(l-n+c) \times (n-c)} \tag{29}$$

Similarly, considering the fundamental incidence matrix $\mathscr{A}$ and the fundamental loop matrix $\mathscr{H}$, their columns follow the order of links. While $\mathscr{A}$ delineates the relationship between nodes and links, $\mathscr{H}$ illustrates the relationship between fundamental loops and links. When a node u_i resides on a specific fundamental loop M_r, it connects to two links e_i and e_j. In $\mathscr{A}$, two columns correspond to this node, implying two elements in the row representing the node u_i. In $\mathscr{H}$, these two links e_i and e_j pertain to the fundamental loop M_r, thus two elements feature in the row representing the fundamental loop M_r.

(1) If both links e_i and e_j align or oppose the direction of the fundamental loop M_r, the corresponding elements in $\mathscr{H}$ are either both 1 and -1. However, one link points towards the node while the other moves away from it. Consequently, the corresponding elements in $\mathscr{A}$ feature 1 and -1;

In Fig. 9a, if $h_{C_r e_i} = h_{C_r e_j} = 1$, then $a_{u_i e_i} = 1$ and $a_{u_i e_j} = -1$. Conversely, if $h_{C_r e_i} = h_{C_r e_j} = -1$, then $a_{u_i e_i} = -1$ and $a_{u_i e_j} = 1$.

(2) In scenarios where one link e_i aligns with the direction and the other e_j opposes the direction of the fundamental loop M_r, the corresponding elements in $\mathscr{H}$ are 1 and -1. Nevertheless, both links either point towards or move away from the node simultaneously. Consequently, the corresponding elements in $\mathscr{A}$ are either both 1 or both -1.

In Fig. 9b, if $a_{u_i e_i} = a_{u_i e_j} = 1$, then $h_{C_r e_i} = 1$ and $h_{C_r e_j} = -1$. Conversely, if $a_{u_i e_i} = a_{u_i e_j} = -1$, then $h_{C_r e_i} = -1$ and $h_{C_r e_j} = 1$. That is, make the product of the corresponding row vectors of matrices $\mathscr{A}$ and $\mathscr{H}$ equal to zero, which is represented as $\mathscr{A}\mathscr{H}^T = 0_{(n-c) \times (l-n+c)}$.

$$\mathscr{H}\mathscr{A}^T = \left[\mathscr{A}\mathscr{H}^T\right]^T = 0_{(l-n+c) \times (n-c)} \tag{30}$$

For the plane matrix $\mathscr{P}$, considering that a plane itself is a special loop, each node can be considered to be on a certain plane. Therefore, the relationships between, the relationship between $\mathscr{A}$, $\mathscr{Q}$, and $\mathscr{H}$ can be directly applied. Thus, $\mathscr{A}\mathscr{P}^T = 0_{(n-c) \times (l-n+c)}$, $\mathscr{P}\mathscr{A}^T = 0_{(l-n+c) \times (n-c)}$, $\mathscr{Q}\mathscr{P}^T = 0_{(n-c) \times (l-n+c)}$, and $\mathscr{P}\mathscr{Q}^T = 0_{(l-n+c) \times (n-c)}$.

$$\begin{cases} \mathscr{A}\mathscr{H}^T = 0_{(n-c)\times(l-n+c)} \\ \mathscr{Q}\mathscr{H}^T = 0_{(n-c)\times(l-n+c)} \\ \mathscr{A}\mathscr{P}^T = 0_{(n-c)\times(l-n+c)} \\ \mathscr{Q}\mathscr{P}^T = 0_{(n-c)\times(l-n+c)} \end{cases}$$

$$\begin{cases} \mathscr{H}\mathscr{A}^T = 0_{(l-n+c)\times(n-c)} \\ \mathscr{H}\mathscr{Q}^T = 0_{(l-n+c)\times(n-c)} \\ \mathscr{P}\mathscr{A}^T = 0_{(l-n+c)\times(n-c)} \\ \mathscr{P}\mathscr{Q}^T = 0_{(l-n+c)\times(n-c)} \end{cases}$$

6.3. Matrix form of node flow conservation

According to the node flow conservation condition in traffic networks, for all ordinary nodes, the total flow entering the node equals the total flow exiting the node, i.e.,

$$\sum_{e\in O_u} v_e = \sum_{e\in D_u} v_e \tag{31}$$

where e is the link, v_e is the traffic flow on the link e, and O_u and D_u are the flow outflow link set and the flow inflow link set at node u, respectively. Existing research often presents the node–link incidence matrix for traffic networks, where $AV = \sum_{e\in O_u} v_e - \sum_{e\in D_u} v_e = 0$, and thoroughly proves its relevant properties.

Theorem 10. *The fundamental incidence matrix* $\mathscr{A} = \{a_{u_i e_j}\}_{(n-c)\times l}$, *the fundamental cut-set matrix* $\mathscr{Q} = \{q_{e_i C_i}\}_{(n-c)\times l}$, *the fundamental loop matrix* $\mathscr{H} = \{h_{e_r m_r}\}_{(l-n+c)\times l}$, *and the plane matrix* $\mathscr{P} = \{p_{e_j m_j}\}_{(l-n+c)\times l}$ *together encapsulate the node flow conservation in traffic networks. These matrices form a comprehensive framework to guarantee that the traffic flows are conserved across the network. Understanding their relationships could provide additional ways for engineers and planners to design and optimize sensor locations for traffic flow observation, as detailed in Table 6 and the associated matrix equations.*

$$\begin{cases} \mathscr{A} = (A_x \quad A_t) \\ \mathscr{Q} = \left(A_t^{-1}A_x \quad I_{n-c}\right) \\ \mathscr{H} = \left(I_{l-n+c} \quad -\left(A_t^{-1}A_x\right)^T\right) \end{cases}$$

$$\begin{cases} H_t = -\left(A_t^{-1}A_x\right)^T \\ Q_f = -H_t^T = A_t^{-1}A_x \end{cases}$$

V represents the vector of flow on all links; V_t represents the vector of branch flows (unobserved link flows); V_x represents the vector of chord flows (fundamental loops), i.e., observed link flows where sensors are deployed; V_{l-n+c} represents the vector of finite planes which may not necessarily be fundamental loops.

Proof: Consider the block matrix:

$$\begin{cases} \mathscr{A} = (A_x \quad A_t) \\ \mathscr{Q} = (Q_f \quad I_{n-c}) \\ \mathscr{H} = (I_{l-n+c} \quad H_t) \end{cases} \tag{32}$$

(1) Substituting $\mathscr{H}\mathscr{Q}^T = 0$, have

Table 6
Matrix analysis node flow conservation equation.

Flow relationship Matrix	$AV = \sum_{e\in O_u} v_e - \sum_{e\in D_u} v_e = 0$	Not observed traffic link flow $V_{no} = V_t$
$\mathscr{A}$	$\mathscr{A}V = 0$	$V_t = -A_t^{-1}A_xV_x$
$\mathscr{Q}$	$\mathscr{Q}V = 0$	$V_t = -Q_fV_x$
$\mathscr{H}$	$\mathscr{H}^TV_x = V$	$V_t = H_t^TV_x$
$\mathscr{P}$	$\mathscr{P}^TV_{l-n+c} = V$	NA

$$\begin{pmatrix} I_{l-n+c} & H_t \end{pmatrix} \begin{pmatrix} Q_f \\ I_{n-c} \end{pmatrix} = Q_f^T + H_t = 0 \tag{33}$$

Thus $H_t = -Q_f^T$ or $Q_f = -H_t^T$, which means the fundamental cut-set matrix and the fundamental loop matrix can be mutually derived.

(2) Similarly, substituting $\mathscr{A}\mathscr{H}^T = 0$, have

$$\begin{pmatrix} A_x & A_t \end{pmatrix} \begin{pmatrix} I_{l-n+c} \\ H_t \end{pmatrix} = A_x + A_t H_t = 0 \tag{34}$$

Since A_t represents the non-singular submatrix associated with branches and nodes, A_t must be invertible.

Then $H_t = -\left(A_t^{-1}A_x\right)^T$, so $Q_f = -H_t^T = A_t^{-1}A_x$, meaning that both the fundamental cut-set matrix and the fundamental loop matrix can be expressed using the fundamental incidence $\mathscr{A}$.

In summary, the fundamental cut-set matrix $\mathscr{Q}$ and the fundamental loop matrix $\mathscr{H}$ can both be represented using the fundamental incidence matrix $\mathscr{A}$.

$$\begin{cases} \mathscr{A} = \begin{pmatrix} A_x & A_t \end{pmatrix} \\ \mathscr{Q} = \begin{pmatrix} A_t^{-1}A_x & I_{n-c} \end{pmatrix} \\ \mathscr{H} = \begin{pmatrix} I_{l-n+c} & -\left(A_t^{-1}A_x\right)^T \end{pmatrix} \end{cases} \tag{35}$$

The fundamental incidence matrix $\mathscr{A}$ can be used to express the flow conservation condition as follows:

$$\mathscr{A}V = 0 \Rightarrow \begin{pmatrix} A_x & A_t \end{pmatrix} \begin{pmatrix} V_x \\ V_t \end{pmatrix} = 0 \Rightarrow A_x V_x + A_t V_t = 0 \Rightarrow V_t = -A_t^{-1}A_x V_x \tag{36}$$

where A_t represents the non-singular submatrix associating branches with nodes, and A_x represents the submatrix associating chords with nodes. Similar to Eq. (9), $V_x = V_{ob}$ (flow on chord links is observed); $V_t = V_{no}$ (flow on branch links is unobserved). This explains He's (2013) proposed deployment strategy for finding a spanning tree of a traffic network by deploying sensors not on branches but on chords.

The node flow conservation condition can also be expressed using the fundamental cut-set matrix $\mathscr{Q}$. By combining its definition and related properties, the flow entering/exiting a cut-set sums up to zero, it can be derived that:

$$\mathscr{Q}V = 0 \Rightarrow \begin{pmatrix} Q_f & I_{n-c} \end{pmatrix} \begin{pmatrix} V_x \\ V_t \end{pmatrix} = 0 \Rightarrow Q_f V_x + V_t = 0 \Rightarrow V_t = -Q_f V_x \tag{37}$$

Combining this with $V_t = -A_t^{-1}A_x V_x$ yields $Q_f = A_t^{-1}A_x$.

Using the fundamental loop matrix $\mathscr{H}$ to express the flow conservation condition. The matrix $\mathscr{H}$ describes the relationship between links and fundamental loops, where fundamental loops consist of branches and one chord.

Therefore, it is important to consider the relationship between all link flows and the flows on the links forming chords. For a link, if it is part of a chord, it is only on one fundamental loop, and the flow on this link is referred to as the loop flow; if it is a branch, then the flow on this link is the sum of the loop flows associated with it, with its sign determined by the relationship between the loop and the link direction. Consequently, the flow on chords represents the loop flow. Each element in a column of matrix $\mathscr{H}$ corresponds exactly to a different loop and has:

$$\mathscr{H}^T V_x = V \Rightarrow \begin{pmatrix} I_{l-n+c} \\ H_t \end{pmatrix} V_x = \begin{pmatrix} V_x \\ V_t \end{pmatrix} \Rightarrow V_t = H_t^T V_x \tag{38}$$

This is the relationship between branch flow and loop flow (chord flow).

The node flow conservation condition can be expressed using plane matrix $\mathscr{P}$. Since planes are special linearly independent loops that may not necessarily be basic, each link's flow can be represented as a sum of plane flows similar to consideration with a fundamental loop matrix.

$$\mathscr{P}^T V_{l-n+c} = V \tag{39}$$

where V_{l-n+c} represents plane flow within loops, but cannot be directly interpreted as individual link flows.

Through the topological properties of graph theory, the fundamental incidence matrix $\mathscr{A} = \left\{a_{u_ie_j}\right\}_{(n-c)\times l}$, the fundamental cut-set matrix $\mathscr{Q} = \left\{q_{e_iC_i}\right\}_{(n-c)\times l}$, the fundamental loop matrix $\mathscr{H} = \left\{h_{e_rm_r}\right\}_{(l-n+c)\times l}$, and the plane matrix $\mathscr{P} = \left\{p_{e_jm_j}\right\}_{(l-n+c)\times l}$ can be respectively linked to the connectivity between links, nodes, fundamental cut-sets, fundamental loops, and planes. Additionally, the proposed framework as shown in the second column of Table 6 provides a new way in the form of matrix equations to describe the node flow conservation condition using passive sensors. Moreover, these matrix equations present new methods to calculate the traffic flows on unobserved links. For example, the matrix equation $V_t = -Q_f V_x$ provides a method for the calculation of unobserved link flows using the fundamental cut-set matrix of traffic network. It is noticed that the matrix equation $V_t = H_t^T V_x$ also yields a new approach for calculating unobserved link flows from the viewpoint of the fundamental loop matrix of traffic networks. Another interesting problem is that the newly proposed equation $V_t = -A_t^{-1} A_x V_x$ provides a theoretical extension of He (2013). He (2013) proposed an algorithm for finding a scheme of senor location using spanning tree. Based on this work, we further propose a matrix-based formula for unobserved link flow calculation using the spanning tree matrix.

Additionally, Theorem 8 further shows that the total number of senor location schemes is $|\mathscr{A}\mathscr{A}^T|$. Theorem 9 describes the virtual network structure of a traffic network, simplifying network analysis through the use of a spanning tree and associated matrices, thereby providing a new theoretical tool for traffic flow relationships. Theorem 10 explains the relationships between the fundamental incidence matrix, the fundamental cut-set matrix, the fundamental loop matrix, and the plane matrix under the condition of node flow conservation, offering a mathematical framework that guarantees traffic flow conservation. In engineering and practical applications, these theorems are designed to help engineers simplify network analysis in a matrix manner, observe the full network traffic flows, and optimize the sensor location schemes (see Section 7.2).

7. NSP metric for optimizing sensor deployment under resource constraints

Existing methods effectively address the problem of achieving full observability by installing passive sensors on links (e.g., Hu et al., 2009; Castillo et al., 2013a, 2014; Ng, 2012; He, 2013) by leveraging the algebraic relationships between link variables to determine the required set of sensors. However, these approaches may be impractical in real-world networks due to their potential need for a large number of sensors. In reality, constraints such as budget limits and a restricted number of sensors lead to insufficient sensor deployment for achieving full observability. Therefore, building upon various solutions for the full observability problem, further research is needed to determine the optimal sensor deployment strategy under resource constraints.

7.1. Assessing partial observability in link flow observability problem

Gentili and Mirchandani (2012) introduced the concept of partial observability, aiming at identifying the minimal subset of the system state variables necessary to achieve a specified level of observability. This concept has been explored further in the works of Castillo et al. (2012) and Ng (2013), focusing on identifying observable links using a predetermined set of sensors. Building on this foundation, Viti et al. (2014) expanded the concept by exploring how observed flows provide information about unobserved variables. Specifically, Viti et al. (2014) developed the concept by examining how observed flows can provide information about unobserved variables. Partial observability is defined as a situation where a variable, which is not directly observed, is still functionally related to at least one of the observed flows. This occurs when, within a fully observable solution, at least one of the observed flows is no longer measured.

Viti et al. (2014) proposed the null space projection (NSP) metric to quantify the quality of solutions under sensor constraints. NSP leverages the null space of the network observation matrix and uses the Frobenius norm to measure errors induced by unobserved variables. It provides a relative score ranging from 0 (fully observable) to 1 (no sensors) for assessing network uncertainty. This approach integrates techniques from fully observable solutions with local search algorithms to minimize error within a specified sensor budget. Specifically, the NSP metric is calculated through the following formula:

$$\mathrm{NSP} = \frac{\| \Omega^T B' \|_F}{\| \Omega^T \|_F} \tag{40}$$

Here, $\Omega = \begin{pmatrix} I & 0 \\ E & 0 \end{pmatrix}$ is the augmented matrix of the link-link relationship matrix E, B' is the basis of the null space of Ω, and $\| \cdot \|_F$ denotes the Frobenius norm.

The link-link relationship matrix E, central to network flow analysis, is derived from the node flow conservation equation. Let E be the link-link relationship matrix of a fully observable solution, as described in the literature (Castillo et al., 2008d; Hu et al., 2009; Ng, 2012; He, 2013). The matrix E is used in the following equations:

$$V_{no} = E V_{ob}, \quad E \in \mathbb{R}^{|no| \times |ob|} \tag{41}$$

where V_{no} represents the vector of linearly related link flows, and V_{ob} represents the vector of linearly independent link flows.

Similarly, for a fully observable solution of link flows, it can be expressed as:

$$V_{dep} = E V_{indep}, \quad E \in \mathbb{R}^{|dep| \times |indep|} \tag{42}$$

In this context, V_{dep} denotes the vector of linearly dependent link flows, and V_{indep} denotes the vector of linearly independent link flows. V_{indep} are observable flow variables measured directly through sensors and form the basis of an observational model. Conversely, V_{dep} includes flow variables that are not directly observable but can be inferred from the network's structure and adherence to flow conservation principles.

The matrix E describing the link-link relationships can be obtained through transformations involving node-link incidence matrices, fundamental cut-set matrices, and fundamental loop matrices. According to Eq. (6) and Theorem 10, matrix E is given by

$$E = -{A_{no}^{*}}^{-1}A_{ob}^{*} = -A_{t}^{-1}A_{x} = -Q_{f} = H_{t}^{T} \tag{43}$$

In the study of partial observability in linear systems, the focus is on the observable matrix E and the augmented matrix $\Omega = \begin{pmatrix} I & 0 \\ E & 0 \end{pmatrix}$. The null space of Ω denoted as $Null(\Omega)$, and its basis is denoted as B'. This null space, along with its basis, is instrumental in quantifying the transformation from unmeasured to observed variables, which is achieved by computing the Frobenius norm $\|\Omega^T B'\|_F$.

To provide a standardized measure, the NSP metric is then relatively normalized to the Frobenius norm of the initial matrix Ω. This normalization process results in the NSP metric formula:

$$\text{NSP} = \frac{\| \Omega^T B'\|_F}{\| \Omega^T\|_F}$$

The NSP metric encapsulates the essence of this transformation, providing a quantifiable and standardized index of the system's degree of observability. It reflects the effectiveness of sensor deployment strategies by revealing the impact of unobserved variables on the overall network's state estimation. This metric can thus be considered as a tool for assessing and optimizing sensor placement in scenarios where full observability may not be feasible, offering valuable information under resource constraints.

7.2. Resource-constrained sensor network optimization (RSNO) model

Under conditions of resource constraints, an optimization model is developed to achieve optimal performance in observing network traffic flows. This model employs the NSP metric as a central tool to quantify and minimize the impact of incomplete observability, caused by budget limitations, on network performance.

- Definition of decision variables:

The decision variable x_{e_i} is defined as a binary variable indicating whether a sensor is deployed on a link e_i within the traffic network. Specifically,

$$x_{e_i} = \begin{cases} 1, & \textit{the installation of a sensor on link } e_i \\ 0 & \textit{indicates no sensor installation} \end{cases}$$

$e_i, i = 1, 2, \ldots, l$ represents all links of the traffic network.

- Establishment of objective function:

The objective of the optimization model is to minimize the performance loss due to the inability to achieve full observability under budget constraints. To this end, the NSP metric is used as the objective function. Such a metric reflects the uncertainty and information loss in network traffic observation given the configuration of deployed sensors. The objective function is formulated as:

$$\textit{Min}\ \text{NSP}(\aleph) = \frac{\| \Omega^{\mathrm{T}} B'\|_F}{\| \Omega^{\mathrm{T}}\|_F} \tag{44}$$

$\aleph$ represents the set of all feasible solutions obtained through matrix-based methods for the link flow observability problem, and NSP ($\aleph$) is the NSP metric value computed based on the sensor deployment scheme.

- The setting of constraints:

(1) **Budget constraint:** The total observation cost must not exceed the predetermined budget C, and the number of sensors installed must not exceed k. This is mathematically expressed as:

$$\sum_{i=1}^{l} c_{e_i} x_{e_i} \leq C; \ \ \sum_{i=1}^{l} x_{e_i} \leq k \tag{45}$$

c_{e_i} is the cost of installing sensors on the link e_i.

(2) **Flow conservation constraint:** According to the principle of flow conservation, the product of the observation matrix Ω and the sensor placement scheme X_i must satisfy the flow matrix equation:

$$\Omega X_i^{\mathrm{T}} = L^{\mathrm{T}} \tag{46}$$

(3) **Feasibility schemes constraint:** The sensor placement scheme $\forall X_i = (x_{e_1}, x_{e_2}, \cdots, x_{e_l})$ must belong to the set of all feasible deployment schemes, which is obtained using matrix-based methods. This is mathematically expressed as:

$$X_i = (x_{e_1}, x_{e_2}, \cdots, x_{e_l}) \in \aleph \tag{47}$$

Using the NSP metric as the objective function, an optimization model is developed under resource constraints to achieve optimal performance in network traffic observation. This model is designated as the RSNO model (resource-constrained sensor network optimization) and is formally described as follows:

$$Min\ \mathrm{NSP}(\aleph) = \frac{\| \Omega^{\mathrm{T}} B' \|_F}{\| \Omega^{\mathrm{T}} \|_F} \tag{48}$$

$$s.t.\ \sum_{i=1}^{l} c_i x_{e_i} \leq C \tag{48a}$$

$$\sum_{i=1}^{l} x_i \leq k \tag{48b}$$

$$\Omega X_i^{\mathrm{T}} = L^{\mathrm{T}} \tag{48c}$$

$$\forall X_i = (x_{e_1}, x_{e_2}, \cdots, x_{e_l}) \in \aleph,\ \aleph = \{X_1, X_2, \dots, X_n\} \tag{48d}$$

It can be seen that the RSNO model (48) is an integer programming problem. If the network scale is not too large, the optimum solution of (48) can be obtained by the enumeration method. This paper adopts the enumeration method to solve the RSNO model (48). The details of the algorithm are omitted due to the length limitation.

7.3. Numerical example

In this section, numerical examples are used to illustrate the application of the resource-constrained sensor network optimization (RSNO) model. Considering a traffic network comprising 5 nodes and 7 links, with nodes 1 and 5 are centroid nodes, as depicted in Fig. 10.

Initially, using the method proposed previously, a sensor deployment scheme for complete observability was determined and shown in Table 7.

Due to budget constraints in practice, it is assumed that only 3 sensors can be deployed within the network. By solving the RSNO model, all possible NSP metric values under resource constraints were obtained and summarized in Table 8.

Table 8 presents the NSP metric values for various sensor deployment schemes within the traffic network depicted in Fig. 10, under specified budget constraints. Through the comparison of these metric values, the optimal sensor deployment scheme can be found with minimal values of the NSP metric. When restricted to the installation of only three sensors, case 8 exhibits the lowest NSP metric value, indicating that the exclusion of a sensor on link 5 in Scheme X_2 has the least impact on the overall network efficiency. The optimal selection for sensor deployment is on links 2, 3, and 4. This deployment strategy minimizes observational error under budgetary constraints, thereby achieving the best observational performance with limited resources.

8. Conclusions

In this paper, by using graph theory and matrix analysis method, the research on the problem of passive sensor deployment on the links to solve the problem of full link flow observability is unified, and the minimum number of sensors required and the number of sensor deployment schemes required are given. The main contributions of this study are summarized as follows:

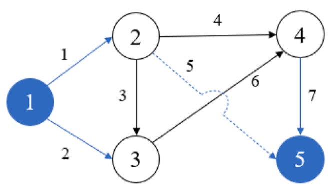


Fig. 10. Traffic network for flow inference.

Table 7
Sensor placement schemes for full observability.

Scheme $\aleph = \{X_1, X_2\}$	Scheme 1	Scheme 2
X_i	$x_{e_3} = x_{e_4} = x_{e_5} = x_{e_6} = 1$ $X_1 = (0\ \ 0\ \ 1\ \ 1\ \ 1\ \ 1\ \ 0)$	$x_{e_2} = x_{e_3} = x_{e_4} = x_{e_5} = 1$ $X_2 = (0\ \ 1\ \ 1\ \ 1\ \ 1\ \ 0\ \ 0)$
Fully observable matrix F	$F = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$	$F = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{pmatrix}$
Augmented matrix Ω:	$\Omega = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$	$\Omega = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$

Table 8
Summary of NSP metrics under resource constraints.

	Links where sensors cannot be installed in fully observable solutions under budget constraints									
Unobserved link	Link 2 $x_{e_2} = 0$		Link 3 $x_{e_3} = 0$		Link 4 $x_{e_4} = 0$		Link 5 $x_{e_5} = 0$		Link 6 $x_{e_6} = 0$	
Case	1	2	3	4	5	6	7	8	9	10
$\aleph$	X_1	X_2	X_1	X_2	X_1	X_2	X_1	X_2	X_1	X_2
NSP	NA	0.3922	0.4097	0.3923	0.4097	0.3922	0.3693	0.3536	0.4264	NA

(1) A virtual network was constructed using an isomorphic graph definition. Conversion between the graph and its virtual network graph was achieved through mutual conversion of their respective node–link incidence matrices while ensuring that both graphs have identical numbers of nodes and links with matching link directions.

(2) Two equivalent formulas were provided based on factors such as number of links, ordinary nodes, centroid nodes, planes, and added links to determine required observed links for achieving full link flow observability in a virtual network without needing to consider type or number of centroid nodes for different traffic networks. This allows us to obtain lower limits for deployment quantities and locations needed for observed links.

(3) When the traffic network has only ordinary nodes (i.e., $c = 0$), if full observability of the link flow is achieved, the number of link flows to be observed is at least the number of links minus the number of nodes plus 1; i.e., the lower bound of the number of observed links flows is $(l - n + 1)$. When the traffic network includes centroid nodes (i.e.,$1 \le c \le n$), if full observability of the link flow is achieved, the number of link flows to be observed is at least the number of links plus the number of centroid nodes minus the number of all nodes; i.e., the lower bound of the number of observed link flows is $(l - n + c)$. When the traffic network is full of centroid nodes (i.e., $c = n$), if full observability of the link flow is achieved, the number of link flows to be observed is at least the number of links in the traffic network; i.e., the lower bound of the observed link flow is l.

(4) The number of finite planes was used to represent the number of link flows to be observed in the traffic network. For the transformed virtual network of a planar traffic network, if full observability of the link flow is achieved, the number of link flows to be observed is at least the number of finite planes in the virtual network; i.e., the lower bound of the number of the observed link flow is h. For the transformed virtual network of a non-planar traffic network, if full observability of the link flow is achieved, the number of link flows to be observed is at least the number of finite planes in the virtual network plus the number of added links; i.e., the lower bound of the number of observed link flows is $(h + k)$.

(5) Through the topological properties of graph theory, connections can be established between the fundamental incidence matrix $\mathscr{A}$, the fundamental cut-set matrix $\mathscr{C}$, the fundamental loop matrix $\mathscr{H}$, and the plane matrix $\mathscr{P}$, and the connectivity of links, nodes, cut-sets, loops, and planes. All these connections are expressed in terms of theorems or propositions with rigorous proofs from the viewpoint of a matrix. Analysis of the node flow conservation equations through graph theory and matrix methods indicates that although the total number of sensors required to achieve full link flow observability is fixed, the spanning tree in the network is not unique as the number of deployment schemes is $|\mathscr{A}\mathscr{A}^T|$ which is usually greater than 1.

(6) The resource-constrained sensor network optimization (RSNO) model has been proposed to optimize the deployment of sensor under incomplete observation due to budget limitations for implementation. By using the null space projection (NSP) metric as its optimization target, the model assesses the impact of incomplete observability on network performance. An enumeration method was used to solve the RSNO model in a small-scale network. Future research needs to be carried out to develop more efficient algorithms for large-scale networks.

CRediT authorship contribution statement

Yue Zhuo: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Hu Shao:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **William H.K. Lam:** Writing – review & editing, Validation, Funding acquisition. **Mei Lam Tam:** Writing – review & editing, Funding acquisition. **Shuhan Cao:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

Anton, H., 1984. Elementary Linear Algebra. John Wiley & Sons, Inc., NY.
Bianco, L., Confessore, G., Reverberi, P., 2001. A network based model for traffic sensor location with implications on O/D matrix estimates. Transp. Sci. 35 (1), 50–60.
Bianco, L., Confessore, G., Gentili, M., 2006. Combinatorial aspects of the sensor location problem. Ann. Oper. Res. 144 (1), 201–234.
Bianco, L., Cerrone, C., Cerulli, R., Gentili, M., 2014. Locating sensors to observe network arc flows: exact and heuristic approaches. Comput. Oper. Res. 46, 12–22.
Bondy, J.A., Murty, U.S.R., 1976. Graph Theory With Applications. Macmillan, London.
Castillo, E., Cobo, A., Jubete, F., Pruneda, R.E., Castillo, C., 2001. An orthogonally based pivoting transformation of matrices and some applications. SIAM J. Matrix Anal. Applic. 22 (3), 666–681.
Castillo, E., Jubete, F., Pruneda, R.E., 2002. Solares C. Obtaining simultaneous solutions of linear subsystems of inequalities and duals. Linear. Algebra Appl. 346 (1–3), 131–154.
Castillo, E., Conejo, A.J., Pruneda, R.E., Solares, C., 2007. Observability in linear systems of equations and inequalities: applications. Comput. Oper. Res. 34 (6), 1708–1720.
Castillo, E., Menéndez, J.M., Sanchez-Cambronero, S., 2008c. Traffic estimation and optimal counting location without path enumeration using Bayesian networks. Comput.-Aided Civ. Infrastruct. Eng. 23 (3), 189–207.
Castillo, E., Menéndez, J.M., Jiménez, P., 2008d. Trip matrix and path flow reconstruction and estimation based on plate scanning and link observations. Transp. Res. Part B 42 (5), 455–481.
Castillo, E., Gallego, I., Menéndez, J.M., Rivas, A., 2010a. Optimal use of plate-scanning resources for route flow estimation in traffic networks. Trans. Intell. Transp. Syst. 11 (2), 380–391.
Castillo, E., Gallego, I., Sánchez-Cambronero, S., Rivas, A., 2010b. Matrix tools for general observability analysis in traffic networks. Trans. Intell. Transp. Syst. 11 (4), 799–813.
Castillo, E., Rivas, A., Jiménez, P., Menéndez, J.M., 2012. Observability in traffic networks. Plate scanning added by counting information. Transportation. (Amst) 39 (6), 1301–1333.
Castillo, E., Calviño, A., Menéndez, J.M., Jiménez, P., Rivas, A., 2013a. Deriving the upper bound of the number of sensors required to know all link flows in a traffic network. Trans. Intell. Transp. Syst. 14 (2), 761–771.
Castillo, E., Nogal, M., Rivas, A., 2013b. Sánchez-Cambronero, S. Observability of traffic networks. Optimal location of counting and scanning devices. Transportmetrica B 1 (1), 68–102.
Castillo, E., Calviño, A., Lo, H.K., Menéndez, J.M., Grande, Z., 2014. Non-planar hole-generated networks and link flow observability based on link counters. Transp. Res. Part B 68, 239–261.
Castillo, E., Grande, Z., Calviño, A., Szeto, W.Y., Lo, H.K., 2015. A state-of-the-art review of the sensor location, flow observability, estimation, and prediction problems in traffic networks. J. Sens. 1–26.
Cerrone, C., Cerulli, R., Gentili, M., 2015. Vehicle-id sensor location for route flow recognition: models and algorithms. Eur. J. Oper. Res. 247 (2), 618–629.
Chin, F., Chrobak, M., Yan, L., 2009. Algorithms for placing monitors in a flow network. In: International Conference on Algorithmic Applications in Management. Springer, Berlin, Heidelberg, pp. 114–128.
Fu, C., Zhu, N., Ling, S., Ma, S., Huang, Y., 2016. Heterogeneous sensor location model for path reconstruction. Transp. Res. Part B 91, 77–97.
Fu, H., Lam, W.H., Shao, H., Kattan, L., Salari, M., 2022. Optimization of multi-type traffic sensor locations for estimation of multi-period origin–destination demands with covariance effects. Transp.Res. Part E 157, 102–555.
Gaudry, M., Lamarre, L., 1979. Estimating origin-destination matrices from traffic counts: a simple linear intercity model for Quebec. Cahiers De Recherche 15.
Gentili, M., Mirchandani, P.B., 2012. Locating sensors on traffic networks: models, challenges and research opportunities. Transp. Res. Part C 24, 227–255.
Gentili, M., Mirchandani, P., 2014. Computational complexity analysis of the sensor location flow observability problem. Optim. Lett. 8, 2245–2259.
Gu, W., Jia, X., 2005. On a traffic control problem. In: 8th International Symposium on Parallel Architectures, Algorithms and Networks (ISPAN'05), 6.
Hadavi, M., Shafahi, Y., 2016. Vehicle identification sensor models for origin–destination estimation. Transp. Res. Part B 89, 82–106.
He, S.X., 2013. A graphical approach to identify sensor locations for link flow inference. Transp. Res. Part B 51, 65–76.
Highway Capacity Manual, 2000. Transportation Research Board. National Research Council, Washington, D.C.
Hu, S., Peeta, S., Chu, C., 2009. Identification of vehicle sensor locations for link-based network. Transp. Res. Part B 43, 873–894.
Hu, S.R., Liou, H.T., 2014. A generalized sensor location model for the estimation of network origin–destination matrices. Transp. Res. Part C 40, 93–110.
Li, X., Ouyang, Y., 2011. Reliable sensor deployment for network traffic surveillance. Transp. Res. Part B 45 (1), 218–231.

Liu, J., Zhou, X., 2019. Observability quantification of public transportation systems with heterogeneous data sources: an information-space projection approach based on discretized space-time network flow models. Transp. Res. Part B 128, 302–323.

Mínguez, R., Sánchez-Cambronero, S., Castillo, E., Jiménez, P., 2010. Optimal traffic plate scanning location for OD trip matrix and route estimation in road networks. Transp. Res. Part B 44 (2), 282–298.

Morrison, D.R., Martonosi, S.E., 2015. Characteristics of optimal solutions to the sensor location problem. Ann. Oper. Res. 226 (1), 463–478.

Ng, M.W., 2012. Synergistic sensor location for link flow inference without path enumeration: a node-based approach. Transp. Res. Part B 46 (6), 781–788.

Ng, M.W., 2013. Partial link flow observability in the presence of initial sensors: solution without path enumeration. Transp. Res. Part E 5, 62–66.

Owais, M., 2022. Traffic sensor location problem: three decades of research. Expert. Syst. Appl., 118134

Salari, M., Kattan, L., Lam, W.H., Lo, H.P., Esfeh, M.A., 2019. Optimization of traffic sensor location for complete link flow observability in traffic network considering sensor failure. Transp. Res. Part B 121, 216–251.

Salari, M., Kattan, L., Lam, W.H., Esfeh, M.A., Fu, H., 2021. Modeling the effect of sensor failure on the location of counting sensors for origin–destination (OD) estimation. Transp. Res. Part C 132, 103–367.

Shan, D., Sun, X., Liu, J., Sun, M., 2018. Optimization of scanning and counting sensor layout for full route observability with a bi-level programming model. Sensors 18 (7), 22–86.

Strang, G., 1986. Introduction to Applied Mathematics. Wellesley-Cambridge Press, Wellesley, MA.

Sun, W., Shao, H., Shen, L., Wu, T., Lam, W.H., Yao, B., Yu, B., 2021. Bi-objective traffic count location model for mean and covariance of origin–destination estimation. Expert. Syst. Appl. 170, 114554.

Sun, W.W., Shao, H., Wu, T., Shao, F., Fainman, E.Z., 2022. Reliable location of automatic vehicle identification sensors to recognize origin–destination demands considering sensor failure. Transp. Res. Part C 136, 103–551.

Viti, F., Corman, F., 2012. A novel approach to the sensor location problem for measuring the observed network flow variability. In: Proceedings of the 5thInternational Symposium of Traffic Network Reliability. Hong Kong, China, 18–19 December.

Viti, F., Rinaldi, M., Corman, F., Tampère, C.M., 2014. Assessing partial observability in network sensor location problems. Transp. Res. Part B 70, 65–89.

Xu, X., Lo, H.K., Chen, A., Castillo, E., 2016. Robust network sensor location for complete link flow observability under uncertainty. Transp. Res. Part B 88, 1–20.

Xu, Z.H., Mo, X.P., Xu, Z.G., Qu, X.B., 2023. A review of road traffic detectors and their optimal layout methods. J. South China Univ. Technol. (Nat. Sci. Ed.) 51 (10).

Yang, H., Zhou, J., 1998. Optimal traffic counting locations for origin-destination matrix estimation. Transp. Res. Part B 32, 109–126.